



Annu. Rev. Plant Biol. 2004. 55:459-93
 doi: 10.1146/annurev.arplant.55.031903.141655
 Copyright © 2004 by Annual Reviews. All rights reserved

HOW DO CROP PLANTS TOLERATE ACID SOILS? MECHANISMS OF ALUMINUM TOLERANCE AND PHOSPHOROUS EFFICIENCY

Leon V. Kochian,¹ Owen A. Hoekenga,²
 and Miguel A. Piñeros³

*U.S. Plant, Soil, and Nutrition Laboratory, USDA-ARS, Cornell University, Ithaca,
 New York 14853; email: 1LVK1@cornell.edu, 2OAH1@cornell.edu,
 3MAP25@cornell.edu*

Key Words manganese tolerance, organic acid exudation, anion channel

■ **Abstract** Acid soils significantly limit crop production worldwide because approximately 50% of the world's potentially arable soils are acidic. Because acid soils are such an important constraint to agriculture, understanding the mechanisms and genes conferring tolerance to acid soil stress has been a focus of intense research interest over the past decade. The primary limitations on acid soils are toxic levels of aluminum (Al) and manganese (Mn), as well as suboptimal levels of phosphorous (P). This review examines our current understanding of the physiological, genetic, and molecular basis for crop Al tolerance, as well as reviews the emerging area of P efficiency, which involves the genetically based ability of some crop genotypes to tolerate P deficiency stress on acid soils. These are interesting times for this field because researchers are on the verge of identifying some of the genes that confer Al tolerance in crop plants; these discoveries will open up new avenues of molecular/physiological inquiry that should greatly advance our understanding of these tolerance mechanisms. Additionally, these breakthroughs will provide new molecular resources for improving crop Al tolerance via both molecular-assisted breeding and biotechnology.

CONTENTS

INTRODUCTION	460
MECHANISMS OF ALUMINUM TOLERANCE	461
Overview of Aluminum Toxicity	461
Physiological Mechanisms of Al Tolerance	461
Other Potential Mechanisms of Al Exclusion	470
Internal Detoxification of Aluminum	471
Tolerance to Toxic Levels of Manganese	472
Genetic Analysis of Aluminum Tolerance	473

Tolerance Loci with Qualitative Inheritance	473
Tolerance Loci with Quantitative Inheritance	475
Genomic Analysis of Al Tolerance	476
PHOSPHOROUS EFFICIENCY: TOLERANCE	
TO P DEFICIENCY STRESS	480
Root Exudates and P Mobilization from the Soil	480
Root Morphology and Architecture	481
Pi Homeostasis and Transport	481
Genetic Analysis of P Efficiency	482
SUMMARY	483

INTRODUCTION

Acid soils, which are soils with a pH of 5.5 or lower, are one of the most important limitations to agricultural production worldwide. Approximately 30% of the world's total land area consists of acid soils, and as much as 50% of the world's potentially arable lands are acidic (159). The production of staple food crops, and in particular grain crops, is negatively impacted by acid soils. For example, 20% of the maize and 13% of the rice production worldwide is on acid soils (159). Furthermore, the tropics and subtropics account for 60% of the acid soils in the world. Thus, acid soils limit crop yields in many developing countries where food production is critical. In developed countries such as the United States, high-input farming practices such as the extensive use of ammonia fertilizers are causing further acidification of agricultural soils. Although liming of acid soils can ameliorate soil acidity, this is neither an economic option for poor farmers nor an effective strategy for alleviating subsoil acidity.

This paper updates our 1995 review on the same topic (67). The previous review heavily emphasized mechanisms of aluminum (Al) toxicity, which significantly impacts plant growth on acid soils. Research on the mechanisms plants employ to tolerate toxic levels of Al in acid soils was in its early stages in 1995. However, over the past eight years, considerable research has focused on physiological and biochemical mechanisms of Al tolerance, as well as the molecular basis for tolerance. Additionally, there has been a growing awareness that there are several factors in addition to Al toxicity that limit crop production on acid soils. This acid soil "syndrome" includes toxic levels of Al, manganese (Mn), and iron (Fe), as well as deficiencies of several essential mineral elements, with phosphorus (P) being the major limiting nutrient on acid soils. Hence, in this review we highlight the key points regarding advances in our understanding of plant responses to acid soils over the past decade. A significant portion of this review focuses on the physiology, biochemistry, and molecular biology of Al tolerance mechanisms. We also address recent research into plant mechanisms of tolerance to phosphorus deficiency stress, a trait that is commonly termed phosphorous (P) efficiency, as well as some recent work on tolerance to Mn toxicity.

MECHANISMS OF ALUMINUM TOLERANCE

Overview of Aluminum Toxicity

Due to space constraints, we focus on mechanisms of Al tolerance and not Al toxicity. The many complex mechanisms by which Al toxicity is manifested are addressed in several reviews (7, 67, 68, 100, 102). However, to place the research on Al tolerance mechanisms in its proper context, it is important to list some of the key features regarding the mechanistic basis for Al toxicity. These include:

- When the soil pH drops below 5, Al^{3+} is solubilized into the soil solution and this is the most important rhizotoxic Al species (63–65).
- The primary symptom of Al toxicity is a rapid (beginning within minutes) inhibition of root growth, resulting in a reduced and damaged root system and limited water and mineral nutrient uptake (see, for example, 7, 59).
- The rapidity of this response indicates that Al first inhibits root cell expansion and elongation; however, over the longer term, cell division is also inhibited (67, 100, 102).
- The site of Al toxicity is localized to the root apex; thus research on tolerance mechanisms also should be focused on this region of the root (138, 148, 150).
- Because Al is so reactive, there are many potential sites including the cell wall, the plasma membrane surface, the cytoskeleton, and the nucleus that could be targets for injury.
- Although most of the root-associated Al is in the apoplast, a small fraction of the Al rapidly enters the symplast and interacts with symplastic targets (75, 146, 155).
- Al disrupts cytoskeletal dynamics, interacts with both microtubules and actin filaments, and could be an important component of Al-induced inhibition of root elongation (14, 40, 151).
- Al interactions with signal transduction pathways, in particular Ca^{2+} homeostasis and signaling, could play a role in toxicity. Al exposure can alter cytosolic Ca^{2+} levels (58, 60, 175), and can interact with and inhibit the enzyme phospholipase C of the phosphoinositide pathway associated with calcium (Ca) signaling (59). For a recent review on this topic, see Reference 129.
- Al exposure elicits the induction of reactive oxygen species (ROS) as well as peroxidative damage to membranes. Although lipid peroxidation is likely not a primary mechanism of toxicity (51, 171), Al-induced ROS generation and associated mitochondrial dysfunction could be involved in Al inhibition of root growth (170).

Physiological Mechanisms of Al Tolerance

Over the past decade, several laboratories around the world have focused their efforts on identifying and characterizing the mechanisms employed by crop plants

that enable them to tolerate toxic levels of Al in acid soils. From this research, it is clear that there are two distinct classes of Al tolerance mechanisms: those that operate to exclude Al from the root apex and those that allow the plant to tolerate Al accumulation in the root and shoot symplasm. Because of the intense interest in this topic, crop Al tolerance has been the subject of many reviews (7, 37, 67, 68, 84, 88, 100). There has been considerable speculation about a number of different Al tolerance mechanisms and it likely that multiple Al tolerance mechanisms are employed by different plant species. However, to date, most of the experimental evidence has focused on root Al exclusion based on Al-activated organic acid (OA) exudation from the root apex. Evidence is also accumulating for a second tolerance mechanism based on internal detoxification of symplastic Al via complexation with organic ligands, again primarily OAs.

Al-ACTIVATED EXUDATION OF ORGANIC ACIDS FROM THE ROOT The first indications in the literature for a root-localized Al exclusion mechanism came from work primarily with wheat, in which it was shown that Al-tolerant genotypes accumulated from three- to eightfold less Al in the root apex (the critical site for Al toxicity) than did Al-sensitive genotypes (see, for example, 18, 134, 157). Significant Al exclusion did not occur in mature root tissues. It was apparent that Al was excluded from both the root cell wall and root symplast, which is consistent with the continuous release of an Al-binding ligand that complexes Al in the rhizosphere and prevents its entry into the root.

The breakthrough was two-part: the demonstration first in snapbean (107) and then in wheat (18, 23) that Al-tolerant genotypes exhibit a strong Al-activated exudation of Al-chelating OAs (citrate in snapbean and malate in wheat) absent in the Al-sensitive genotypes. Delhaize et al.'s work in wheat provides the most detailed characterization of this system (18, 23, 136, 137). They showed that the Al-activated malate release is localized very specifically to the first few millimeters of the root apex. Also, the release is activated very rapidly by Al, within minutes, indicating that all of the machinery for the malate release is in place in root cells prior to Al exposure, and activation occurs at the protein, not gene, level. Presumably, a continuous exudation of malate increases the malate concentration in the unstirred layer at the surface of the root apex to levels high enough to chelate and detoxify a significant fraction of the rhizosphere Al in contact with the root tip, preventing Al entry into the root. The OA exudation continues as the root grows through the soil, thus maintaining the Al-chelating barrier around the root apex as it encounters new regions of acid soil.

Since these initial reports, high levels of Al-activated OAs have been correlated with differential Al tolerance in a large number of both monocot and dicot plant species, as summarized in Table 1. When the extensive literature on Al-activated organic release from roots of Al-tolerant genotypes is examined as a whole, one can make a strong case that this process is a bona fide Al tolerance mechanism in plants. The supporting evidence can be summarized as follows:

TABLE 1 Plant species where Al-activated root organic acid exudation is correlated with Al tolerance

Plant species (common name)	Genotype	Organic acid released	References
Triticum aestivum (wheat)	Line ET3	Malate	(18, 23, 135, 136)
	Atlas 66	Malate	(53, 122)
	Chinese Spring and derived ditelosomic lines	Malate	(120)
	Kitakami B		(55)
Zea mays (maize)	SA3	Citrate	(121)
	IAC-TAIUBA	Citrate	(61)
	Cateto-Colombia	Citrate	(125)
	ATP-Y	Citrate	(70)
	Sikuani	Citrate, oxalate	(62)
	DK789	Citrate	(55)
Avena sativa (oat)		Citrate, malate	(177)
Secale cereale (rye)		Citrate, malate	(78)
Sorghum bicolor (sorghum)	SC283 and derived NILs	Citrate	(98)
Triticale ssp (triticale)		Citrate, malate	(90)
Colocasia esculenta (taro)		Oxalate	(94)
Nicotiana tabacum (tobacco)		Citrate	(20)
Cassia tora (sickle senna)		Citrate	(55, 91)
Glycine max (soybean)	PI 41,6937	Citrate	(147)
	Suzunari	Citrate	(173)
Helianthus annuus (sunflower)		Citrate, malate	(140)
Fagopyrum esculentum (buckwheat)		Oxalate	(92, 177, 178)
Raphanus sativus (radish)		Citrate, malate	(177)
Brassica napus (rape)		Citrate, malate	(177)
Arabidopsis thaliana		Malate	(50)

- There is a strong correlation between Al tolerance and Al-activated OA release in numerous plant species (Table 1).
- The addition of OAs (malate, citrate, or oxalate) to root bathing solutions ameliorates Al toxicity in Al-sensitive varieties (for example, see 23, 122, 178).
- Complexes of Al with di- and tricarboxylic OAs are not transported across membranes or absorbed by roots (1; D. Jones & L. Kochian, unpublished results).

- Al-activated malate release cosegregates with Al tolerance and Al exclusion from the wheat root apex in near-isogenic lines that differ at a single locus for Al tolerance (18, 23).
- The Al-activated OS's exudation is localized to the root apex, which is the primary site for Al toxicity (138, 150).
- In general, the activation of this mechanism is triggered specifically by exogenous Al^{3+} (136), although some lanthanide cations can act as Al^{3+} analogs in triggering the exudation response.
- The loss of specific chromosome arms that result in a decrease in Al tolerance in ditelosomic wheat lines correlates closely with decreased rates of root apical malate release and decreased Al exclusion from the root apex (120).
- Overexpression of enzymes involved in OA metabolism (citrate synthase and malate dehydrogenase) in transgenic tobacco, *Arabidopsis*, and alfalfa can, in some cases, result in increased root OA content and exudation, as well as enhanced Al tolerance (17, 71, 156).
- D) An Al-gated anion channel was identified in protoplasts isolated from the root apex of Al-tolerant wheat and maize; it is a candidate for the transport system facilitating Al-induced OA release in these species (124, 125, 139, 176).

Although Al-tolerant genotypes from many plant species share this general tolerance mechanism, there are several interesting species-specific differences in this Al-activated response. First, although Al-activated malate exudation has been the most extensively characterized Al-exclusion mechanism (primarily due to the many studies in wheat), a number of plant species including maize (61, 121), sorghum (98), oat (177), radish (177), soybean (147, 173), and tobacco (20) employ Al-activated citrate exudation, whereas buckwheat and taro depend on root oxalate release (94, 178). Some species, such as rye, triticale, and oilseed rape, exhibit an Al-activated exudation of both citrate and malate (78, 90, 177). The significance of the large number of species exhibiting Al-activated citrate release is that citrate³⁻, as the tricarboxylic acid anion, chelates Al^{3+} much more strongly than does the dicarboxylic malate²⁻ anion (formation constant for Al:citrate of 9.6 compared to 5.7 for Al:malate). Hence, citrate is much more effective at detoxifying Al than is malate, and there are distinct advantages to employing citrate exudation to exclude Al, compared with malate. One question for future research is how different plant species regulate the specificity of their OA exudation response. Are species-dependent differences in the OA species based on differences in plasma membrane transporter selectivity and permeability, OA metabolism and compartmentation, or a combination of these factors?

In addition to the identity of the OA released by a specific crop species, there also are differences in the spatial localization of this response in certain species. In wheat, the Al-activated malate exudation is highly localized, occurring in the terminal 0.2–0.3 mm of root. This should increase the efficiency of the response, as exudation that is localized to the specific root zone requiring protection from Al

should minimize the metabolic cost with regard to OA synthesis and loss. However, in maize and sorghum, Al-activated citrate release occurs over much more of the root. In maize, the Al-dependent citrate efflux occurred over the entire terminal 5 cm of root, and both stellar and cortical root cells mediated this response (125), whereas in sorghum, both Al exclusion and the Al-activated citrate release occurred as far back as 3 cm from the root tip (98; J. Magalhaes, J. Shaff & L. Kochian, unpublished results). The site of Al-induced injury in maize roots is localized to the first 3 mm of the root (150), so it appears that in these species there is significant citrate exudation from the mature root that may not be directly involved in Al tolerance.

IS THIS TOLERANCE MECHANISM AL-INDUCIBLE? An important issue that won't be resolved until the genes conferring Al tolerance are isolated and characterized is whether one or more Al-tolerance mechanisms are inducible at the gene expression level. The assumption that Al-tolerance genes and proteins are inducible by Al exposure has been the driving force for many of the molecular studies detailed below. Because this is a field in which our understanding of the physiological mechanisms has preceded our knowledge of the molecular basis for Al tolerance, it is worthwhile to examine the concept of Al tolerance induction from the physiological point of view. As Ma and colleagues pointed out (84, 88), there are two temporal patterns for Al-activated OA exudation. One pattern is represented by plant species such as wheat, where root malate exudation is rapidly activated by Al exposure and there isn't any discernible increase in the rate of malate efflux over time. Therefore, in species like wheat Al activates an already expressed OA transporter, and gene activation does not play a role. However, in plant species such as rye, triticale, and *Cassia tora* (sickle senna), there is a discernible lag in Al-activated OA exudation, and the rate of exudation increases over the first 12–24 h of Al exposure (78, 91). Including maize in this second "Al-inducible" group is somewhat problematic. Our earlier work on Al-activated citrate release in the Al-tolerant cultivar South American 3 shows that the rate of exudation in Al increased significantly over time (121). A similar inducible pattern for root citrate exudation also occurred in a study of a second Al-tolerant maize genotype (61). However, in a more recent study with the very Al-tolerant Brazilian hybrid Cateto-Colombia, there was no evidence for an increased rate of citrate exudation over a four-day exposure to Al (125).

The pattern of increasing rates of OA exudation with longer Al exposures times is consistent with induction of Al tolerance genes contributing to this increased capacity. The question remains, what is being induced? Possible candidates include the increased synthesis or activity (via expression of a second regulatory protein) of a plasma membrane transporter, as well as enzymes involved in OA synthesis. Al could induce alterations in internal OA compartmentation such that more OA is available for transport. To date, no strong evidence has been presented implicating changes in key enzymes catalyzing OA synthesis and metabolism [phosphoenolpyruvate (PEP) carboxylase, malate dehydrogenase, citrate synthase, isocitrate

dehydrogenase] in this inductive response. In rye and triticale, Al activates an up to tenfold increase in citrate and malate exudation with little or no change in the activities of PEP carboxylase, isocitrate dehydrogenase, malate dehydrogenase, and citrate synthase in the root tips of both Al-tolerant and -sensitive cultivars (45, 78).

If Al induces an OA exudation process whose rate increases over time, this should also induce a measurable increase in Al tolerance. It is surprising that this has not been addressed in plant species such as rye where there is a noticeable induction and time-dependent increase in OA release. In sorghum, Al-activated root citrate exudation correlates closely with Al tolerance. When the roots of an Al-tolerant near-isogenic line of sorghum were exposed to a moderate level of Al for a prolonged period of time (six days), there was a significant increase in Al tolerance (98; J. Magalhaes & L. Kochian, unpublished results). That is, after 24–48 h of Al exposure, root growth was inhibited about 40%, whereas by day six in Al, there was no inhibition of root growth. Over the same time period, the Al-activated root citrate exudation actually exhibited a slight decrease, suggesting that another process was induced to facilitate this increase in Al tolerance.

Cellular mechanisms of organic acid transport Because Al exposure rapidly activates the exudation of OAs, and the release is specific for one or two OAs from a cytoplasm that contains a number of different OAs, it is likely that Al activation of a specific plasma membrane transporter is a significant component of this tolerance response. In wheat, Al activates malate release almost instantly, suggesting that transport is the limiting step (119, 136). Further support for this comes from work where no differences in root tip malate content or in the activities of PEP carboxylase or malate dehydrogenase in Al-tolerant vs. -sensitive wheat root tips were observed even though Al exposure activated a large and continuous efflux of malate in the Al-tolerant genotype (23, 136).

Al-ACTIVATED ANION CHANNELS In the root cell cytoplasm with a pH of near neutrality, OAs are almost completely dissociated and thus exist as OA anions. Primarily because of the large negative-inside transmembrane electrical potential in plant cells, there is a strong thermodynamic gradient directed out of the cell for these anions. Therefore, a likely transport mechanism that could mediate Al-activated OA exudation would be an anion channel that opens upon exposure to Al, allowing the OA anion to flow out of the cytoplasm. Anion channels that are specifically activated by extracellular Al^{3+} were recently identified using the patch clamp technique with protoplasts isolated from root tips of Al-tolerant wheat (139, 176) and maize (70, 124, 125). In wheat, Ryan and coworkers (139) used the patch clamp technique in the whole-cell configuration to show that Al^{3+} activated an inward Cl^- current (Cl^- efflux; Cl^- was the only anion in the patch pipette) across the plasma membrane of wheat root protoplasts from the Al-tolerant line; as long as Al was maintained in the external solution, this channel remained open. This channel exhibited transport properties similar to that of Al-activated malate

exudation in intact wheat roots, suggesting that it is mediating the OA exudation in wheat. In a subsequent study from the same group, they showed that the channel was more permeable to malate²⁻ anions than Cl⁻ (176). They also found significant differences in the Al-activated anion channel activity in root tip protoplasts from Al-tolerant versus -sensitive near-isogenic lines; Al³⁺ activated the anion current more frequently and the anion current density was considerably higher in cells from the Al-tolerant line. These findings suggest that root tip cells of both Al-tolerant and -sensitive wheat have this transport system, but it is expressed to a higher level and/or is more active in the tolerant genotype.

In an Al-tolerant maize line that exhibits Al-activated root citrate release, Al³⁺ activated an anion channel that mediates Cl⁻ efflux in root tip protoplasts (124, 125). This Al-activated anion channel exhibited similar properties to the citrate exudation observed in intact maize roots. The key finding from this study was the observation that this channel could be activated in outside-out excised membrane patches, where the functioning of a single channel in the membrane patch in isolation from cytosolic factors can be studied. As Figure 1 (see color insert) shows, excised membrane patches isolated in the absence of Al were electrically quiet. When these membrane patches were exposed to extracellular Al³⁺, the inward anion current indicative of anion efflux was activated, and then the channel closed again when Al was removed. This finding is significant because all the machinery for Al activation of OA exudation is either contained within the channel protein itself or via an associated membrane receptor. As depicted in the model in Figure 2 (see color insert), and described in detail in the review of root OA exudation by Ryan et al. (135), there are three possible scenarios for Al activation of a plasma membrane anion channel involved in OA exudation: (a) The Al³⁺ ion could directly bind to and activate the channel; (b) it could bind to a separate but closely associated membrane receptor which in turn, activates the channel; or (c) Al³⁺ could activate the channel indirectly through a signal cascade that might involve cytosolic components. The findings in Al-tolerant maize suggest that either option (a) or (b) is the most likely scenario. Findings such as these are helping researchers develop molecular strategies for isolating Al tolerance genes in crop plants.

In the aforementioned study in maize, it was not determined whether the channel was permeable to citrate or other OAs because Cl⁻ was used as the primary transported anion. However, in another patch clamp study on Al-activated anion channels in protoplasts from Al-tolerant maize roots, the Al-activated anion channel was permeable to Cl⁻, malate²⁻, and citrate³⁻ (70).

The physiological and electrophysiological studies described above for Al-activated OA transport indicate that an anion channel may be the protein that mediates the OA anion transmembrane flux. This raises the question as to the nature of the anion channel protein. There are several families of anion channels that have been identified, with much of the work coming from animal studies. These include the CLC (chloride channel) family and a subset of the ATP-binding cassette (ABC) protein superfamily. ABC proteins are transporters that bind ATP

during the transport of a wide range of organic and inorganic solutes. A subset of this family includes the cystic fibrosis transmembrane regulator in mammalian cell membranes, which is a Cl^- channel (3). Another ABC protein, Pdr12, is involved in the efflux of OAs in yeast (126). Therefore, could the Al-activated malate transport in wheat be mediated by an ABC transporter? This channel shares several transport similarities with an anion channel in the guard cell plasma membrane, the “slow” anion channel (for slow inactivation), which also mediates the sustained release of anions (76). Based on similarities to the CFTR Cl^- channel, it has been suggested that the slow anion channel in guard cells could be an ABC transporter (76). One similarity is sensitivity to the ABC transporter antagonist, diphenylamine-2-carboxylic acid (DPC). In the electrophysiological investigation of the Al-activated anion channel in Al-tolerant wheat roots described above (176), DPC inhibited this anion channel, and also inhibited the Al-activated malate efflux from intact wheat roots.

Having speculated that this malate transporter may be an ABC protein, some recent work suggests it actually may be a novel type of membrane transporter. A potentially exciting recent finding regarding OA transporters involved in Al tolerance comes from work in Matsumoto’s laboratory describing the isolation of a wheat gene that encodes the root tip Al-activated malate transporter (142). This gene was identified via a subtractive hybridization approach using a pair of near-isogenic wheat lines differing at a single Al tolerance locus. When genes from this screen were further analyzed to find those that were expressed more strongly in the root tip of the Al-tolerant near-isogenic line, a gene that appears to be a novel membrane protein was identified. Heterologous expression of this gene, named *ALMT1* for Al-activated malate transporter, conferred an Al-activated malate exudation in *Xenopus* oocytes, as well as in roots of transgenic rice seedlings and tobacco suspension cells. Expression of *ALMT1* increased Al tolerance in the tobacco suspension cells. This is a potentially important discovery because it may represent the identification of the first major Al tolerance gene in crop plants, and it certainly opens up new avenues of research in this field.

HOW ARE THESE ORGANIC ACID TRANSPORTERS ACTIVATED BY Al? The results from these biophysical and physiological studies clearly indicate that perception of the Al signal and transduction of this signal into activation of OA exudation are important components of the tolerance response. This is an area for which almost no information exists and should be a focus of future research. Based on what is known about ligand-gated channels in general, there are many potential modulators including proteins that could vary in activity or abundance, such as kinases, phosphatases, G proteins, and 14–3–3 proteins, as well as soluble factors that could likewise vary in activity or abundance, such in Ca^{2+} or cyclic nucleotides. Evidence for the possible involvement of kinase-mediated protein phosphorylation of the wheat malate transporter comes from the work of Osawa & Matsumoto (119). They showed that the rapid Al activation of malate release from the root tip of an Al-tolerant wheat genotype (detected within 5 min

after Al exposure) was blocked by the protein kinase inhibitor K-252a. This same treatment with K-252a also abolished Al exclusion and Al tolerance. Al exposure very rapidly, within 30 sec, activated a 48-kD protein kinase from the root tip and this activation was transient, diminishing after 5 min of Al exposure. These findings suggest that protein phosphorylation may be required for activation of this malate transporter by Al. The recent identification of ALMT1, a candidate for the Al-activated malate transporter in wheat (142), opens up new avenues of research into the identification of proteins or other factors that could be directly or indirectly interacting with the OA transporter during Al-mediated activation.

Recently, a member of the WAK family of cell wall-associated protein kinases in *Arabidopsis thaliana*, *WAK1*, was shown to be rapidly (in less than 3 hrs) induced by Al exposure in *Arabidopsis* roots (149). These proteins are predicted to contain several motifs, including an extracellular domain covalently linked to pectin moieties in the cell wall, a transmembrane domain, and a conserved Ser/Thr kinase in a cytoplasmic domain (2). Hence, these proteins have all the structural features to perceive an Al signal in the cell wall and transduce this signal in the plasma membrane or cytosol. Sivaguru and colleagues found the protein to be expressed in the cells of the root periphery of the root tip, and overexpression of *WAK1* in transgenic *Arabidopsis* conferred a small increase in Al tolerance. Although the connection between WAK and Al tolerance from this study is still tenuous, it deserves further examination.

THE PRESENCE OF OTHER TOLERANCE MECHANISMS IN PLANTS EXHIBITING Al-ACTIVATED ORGANIC ACID RELEASE Given the frequent speculation that a number of different tolerance mechanisms could be operating in plants, it is not surprising that researchers are starting to see evidence for additional Al tolerance mechanisms in plant species that depend on Al-activated OA exudation. The best example of this is in buckwheat, where Al exclusion from the root apex due to Al-activated oxalate exudation is coupled with a second, internal detoxification mechanism involving Al chelation with oxalate in the leaves and citrate in the xylem (85, 86, 178). Additional evidence for multiple tolerance mechanisms comes from analyses of the genetic complexity of Al tolerance in rice and maize, where it is a quantitative trait with multiple contributing loci. This suggests that in rice, which is the most Al tolerant of the small grain cereal crops (89), and maize, several different Al tolerance mechanisms could combine and thus result in the high level of observed tolerance. In maize, it is generally accepted that Al tolerance is conferred by Al-activated citrate exudation (61, 121). In both of these studies, this assertion is based on the correlation of rates of Al-activated citrate exudation with Al tolerance between a single Al-tolerant genotype and one or two Al-sensitive lines. Recently, we conducted a similar analysis with a broader panel of Al-tolerant and -sensitive maize lines from Brazil and North America (123). All of the very Al-tolerant Brazilian lines exhibited a high rate of Al-activated citrate release, as expected. However, several of the Al-sensitive lines from Brazil and North America also

exhibited high rates of Al-activated citrate release. In fact, the North American maize inbred, Mo17, exhibited the highest rate of citrate exudation even though it is relatively Al sensitive. Our interpretation of this is that in maize, citrate release is only one of several possible mechanisms operating simultaneously in the very tolerant genotypes, whereas in Mo17, it is the only mechanism functioning. Another North American maize inbred that exhibits a similarly low level of Al tolerance, B73, has no Al-activated citrate release and thus must employ a different mechanism for its basal level of tolerance. Quantitative trait locus (QTL) analysis for Al tolerance with a set of recombinant inbred lines (RILs) generated from a cross of B73 × Mo17 shows that some of the RILs are considerably more Al tolerant than is either parent (49). The existence of transgressive segregation is consistent with some of the progeny combining the different Al tolerance mechanisms we speculate are operating in Mo17 and B73.

What other Al tolerance mechanisms could be operating in these species? With regard to other possible exclusion mechanisms, these could include the root exudation of other Al-chelating organic ligands, the formation of a rhizosphere pH barrier, Al-binding by mucilage secreted from the roots, or the actual efflux of accumulated Al from the root apex via some type of Al transporter. Internal detoxification mechanisms could include Al fixation in the cell wall, complexation in the symplasm via organic ligands, and sequestration in the vacuole.

Other Potential Mechanisms of Al Exclusion

ROOT EXUDATION OF PHENOLIC COMPOUNDS Because of the ability of phenolic compounds to complex metals such as Al and also act as strong antioxidants in response to abiotic stress, they are beginning to receive attention in relation to Al tolerance (101, 118). Phenolics, which are characterized as organic compounds containing one or more hydroxylated aromatic rings, represent a broad range of plant compounds, including alkaloids, flavonoids, terpenoids, and glycosides. They reportedly form strong complexes with Al^{3+} at neutral pH and were implicated in internal Al detoxification in tea and other Al-accumulating species (101, 118). However, the role of root phenolic exudation as an Al exclusion mechanism in the rhizosphere is less clear. The reason for this is that in acidic conditions, Al^{3+} and H^+ compete for the binding sites in phenolic compounds, thus reducing their metal-binding capabilities when compared to most OAs in acidic solutions. Nevertheless, their potential role in Al exclusion cannot be disregarded. Recently, Kidd et al. (62) reported a better correlation between the rate of Al-stimulated root exudation of the flavonoids catechin and quercetin, and differential Al tolerance in three maize genotypes, than with Al-activated OA exudation. The authors suggested that Al-activated exudation of phenolics may play an important role in the detoxification of Al in the rhizosphere surrounding the root apex.

ROOT-MEDIATED INCREASE IN RHIZOSPHERE pH An attractive mechanism that was the topic of early speculation in the Al tolerance literature involved root-mediated

increases in rhizosphere pH. If the root could effectively increase the pH of the rhizosphere surrounding the root tip, this would reduce the activity of the rhizotoxic Al^{3+} species in favor of less toxic hydroxyl monomers of Al. For a review of the earlier literature on this topic, please see Reference 154. Despite the attractiveness of this process as an Al tolerance mechanism, only one publication unequivocally demonstrated a role in Al tolerance. In a study where two classes of Al-tolerant *Arabidopsis* mutants were isolated and characterized, it was shown that Al tolerance in both classes was mediated by Al exclusion (74). One class of mutants exhibited an increased root malate and citrate exudation, whereas the other class did not release any more OA than wild-type seedlings. A companion study showed that in this second class of mutants, Al tolerance was correlated with an Al-activated root apical H^+ influx, measured with a vibrating pH microelectrode system (16). This H^+ influx resulted in an increase in rhizosphere pH at the surface of the root apex, which was large enough to significantly decrease the Al^{3+} activity around the root tip and lead to improved root growth.

Internal Detoxification of Aluminum

Recently, some researchers have turned their attention to plant species that can accumulate Al to high levels in the shoot, to look for mechanisms of internal Al detoxification. The work of Ma and colleagues has begun to provide insights into mechanisms of internal Al detoxification in two Al accumulator species, *Hydrangea* and buckwheat. *Hydrangea* is an ornamental plant whose flowers (actually the sepals) turn from red to blue when the soil is acidified; this color change is due to Al accumulation in the sepals and the formation of a blue colored complex of Al, delphinidin-3-glucoside, and 3-caffeoylquinic acid. *Hydrangea* can accumulate greater than 3000 ppm Al in its leaves (87) and the authors used ^{27}Al -NMR spectroscopy to show that the Al in the leaves exists primarily as a 1:1 Al-citrate complex. In a cytosol with a pH of around 7, this would be a very strong complex (higher stability constant than for Al-ATP) and should protect the cytosol against Al injury. In another series of investigations with the Al-tolerant crop species buckwheat, Ma and colleagues (178) showed that part of its Al tolerance involves Al-activated oxalate exudation from the root apex. Buckwheat also accumulates Al as high as 15,000 ppm in leaves when grown on acid soils (88). The authors showed that in the roots and leaves, most of the Al is complexed with oxalate in a 1:3 Al:oxalate complex (86). It was later found that Al in the xylem stream is complexed with citrate, not oxalate (85). Hence, it appears that the Al undergoes a ligand exchange from oxalate to citrate when it is transported into the xylem, and is exchanged back into oxalate when transported into the leaves. In another study, protoplasts and vacuoles were isolated from buckwheat leaves that had accumulated Al. More than 80% of the leaf Al was in the leaf protoplasts as a 1:3 Al:oxalate complex, and most of this Al:oxalate complex was sequestered in the vacuole (144). Therefore, as the model in Figure 2 depicts, this mechanism involves Al chelation in the cytosol and subsequent storage of the Al-OA complex in the vacuole.

Most studies on mechanisms of Al tolerance have focused on Al-tolerant genotypes of crop species with a long history of breeding and cultivation. However, recently researchers have begun looking at a forage grass known as signalgrass (*Brachiaria decumbens*), which is widely sown in the tropics and is derived from wild germplasm. Signalgrass is extremely Al tolerant, and is being studied to determine if it possesses novel mechanisms of tolerance. In a hydroponic study comparing signalgrass with a less Al-tolerant relative, ruzigrass (*Brachiaria ruziziensis*), signalgrass was considerably more Al tolerant than tolerant genotypes of wheat, maize, and triticale (164). Some degree of Al exclusion could be operating in signalgrass roots, as ruzigrass accumulates more than twofold higher Al in its root tips compared with signal grass. No evidence for Al activation of root OA exudation or root-mediated alteration in rhizosphere pH was found. In a subsequent study, internal Al detoxification in the root was investigated in the two forage grasses (163). Al exposure triggered a significant increase in citrate, and to a lesser degree trans-aconitate, in the root tips of both *Brachiaria* species, and only a very small fraction of these OAs was secreted from the root tips. The authors suggested that internal detoxification and sequestration of Al with citrate could play a role in the high basal level of Al tolerance in both species, but could not explain the much higher level of Al tolerance in signalgrass. This Al-tolerant species may turn out to be an important source of novel Al tolerance mechanisms and genes, and many laboratories are initiating programs to investigate the genetic and molecular basis of Al tolerance in *Brachiaria decumbens*.

Tolerance to Toxic Levels of Manganese

Many acid soils contain excessively high levels of both Mn and Al; Mn toxicity is possibly the second most important metal toxicity on acid soils after Al (32, 33). Whereas Al toxicity is primarily expressed in the root, Mn toxicity symptoms are localized to the shoot, and are characterized by stunted growth, chlorosis, and necrotic lesions in the leaves. Although the physiological mechanisms underlying Mn toxicity and tolerance have not yet been elucidated, several studies suggest that Mn-induced oxidative stress is a cause of Mn toxicity. Additionally, differences in the ability to ameliorate this Mn-induced oxidative stress could be involved in the genetically based variability in Mn tolerance seen in many crop species (see, for example, 39, 52).

One research area that may have relevance to the field of Mn tolerance involves the recent identification of several plant genes that encode transporters with the ability to transport Mn^{2+} . Hirschi et al. (48) isolated and characterized the *Arabidopsis* antiporter, CAX2, which is a homolog of a previously identified tonoplast Ca^{2+}/H^{+} antiporter, CAX1. They showed that the CAX2 protein was localized to the tonoplast, and expression of CAX2 in transgenic tobacco conferred increases in Mn tolerance as well as Mn accumulation in both whole plants and isolated tonoplast vesicles. They hypothesized that CAX2 might be a broad substrate range divalent cation transporter that could play a role in Mn sequestration in the

vacuole and Mn tolerance. In another study, Wu et al. (169) found that a member of the Ca-ATPase subgroup of the P-type ATPase family, ECA1, which was previously an endoplasmic reticulum-localized Ca-ATPase (79), also could transport Mn^{2+} ions. *ECA1* knockouts in *Arabidopsis* were Mn sensitive, and when *ECA1* was expressed in yeast, it conferred Mn tolerance. Wu and colleagues (169) speculated that ECA1 participates in both Ca and Mn homeostasis. Very recently, Delhaize et al. (21) isolated another putative Mn transporter gene, *ShMTP1*, which is a member of yet a third family of metal transporters, the cation diffusion facilitator family. *ShMTP1* was cloned via functional complementation in yeast using a cDNA library constructed from a tropical legume, *Stylosanthes hamata*, which is tolerant to both acid soils and high levels of Mn. When expressed in yeast, this transporter conferred Mn tolerance via internal sequestration and functioned as an endomembrane Mn^{2+}/H^{+} antiporter. When expressed in *Arabidopsis*, it conferred increased Mn tolerance as well as enhanced shoot Mn accumulation. Expression of a ShMTP1-GFP chimeric protein showed that ShMTP1 is localized to the *Arabidopsis* tonoplast membrane and presumably participates in Mn homeostasis by sequestering this essential but potentially toxic metal in the vacuole. These three studies illustrate the potential for identifying metal transporter genes that possibly play a role in the natural variation in Mn tolerance seen in plants, as well as in providing new molecular tools that possibly can be used to generate transgenic plants with superior Mn tolerance on acid soils.

Genetic Analysis of Aluminum Tolerance

Understanding the inheritance of Al tolerance has been an active area of research for many years, driven largely by work in the plant breeding community as increasing tolerance to Al toxicity is a priority for increasing production on acid soils (see 37, 46 for recent reviews). The majority of work has been performed in cereals, especially among members of the Triticeae (e.g., wheat, rye). This is perhaps due to a combination of national priority/geography and the simple inheritance of tolerance in that tribe. In spite of this body of work, no bona fide Al tolerance genes have yet been cloned from any species (but see 142). Recently, research was conducted on species (e.g., rice, *Arabidopsis*) where Al tolerance behaves as a quantitative trait. Despite the difficulties associated with the analysis of quantitative traits, the existence of sequenced and annotated genomes for rice and *Arabidopsis* should greatly accelerate the rate of discovery for genes that underlie this trait.

Tolerance Loci with Qualitative Inheritance

Wheat is the best-characterized genetic system for analyzing Al tolerance. Crop improvement programs in Brazil and the United States have lead to the development of excellent cultivars (cv. BH1146 and Atlas66, respectively), which have been subsequently well studied in both field and laboratory conditions. In many crosses between these elite cultivars and Al-sensitive varieties, Al tolerance segregates as a single, dominant locus; in other crosses, segregation patterns suggest

that two loci are responsible for tolerance (37). One of these loci has been mapped to the long arm of chromosome 4D, sometimes called *Alt_{BH}* or *Alt2* (81, 133). The existence of other loci in the genome can further be inferred from the study of varieties that contain chromosomal deletions (e.g., nullisomics, ditelosomics, or small segmental deletions) and have diminished tolerance to Al stress (4, 120). However, the importance of these additional locations beyond 4DL (e.g., 5AS, 7AS) has been difficult to evaluate because genetic experiments have yet to implicate these regions as containing tolerance genes by cosegregation analysis.

As mentioned previously, Al tolerance in wheat is highly correlated with an Al-activated release of malate. In a survey of 36 cultivars, Ryan and coworkers (137) quantified both relative root length (RRL) and malate release to estimate Al tolerance. A correlation analysis of these two variables demonstrated that 84% of differences observed in RRL were explained by the quantity of malate released. This suggests that the majority of variance in Al tolerance observed between wheat cultivars exists within a single physiological mechanism. Taken together with the observations regarding cosegregation analysis, it is possible that a few gatekeeper loci are responsible for the differences in Al tolerance reported in wheat.

Rye (*Secale cereale*), a close relative of wheat, is generally regarded as having superior tolerance to abiotic stresses, including Al (4). Unlike wheat, rye is self-incompatible and thus an obligate out-crossing species. This may help explain why cosegregation experiments in rye generally detect a greater number of Al tolerance loci than are detected in wheat (4, 35, 46). Like wheat, the long arm of chromosome 4 contains a major Al tolerance locus, *Alt3* (35). An improved map estimate for *Alt3* demonstrates tight linkage with markers linked to *Alt_{BH}*, advancing the suggestion that homeologous loci act as Al tolerance genes in both species (104). A second tolerance locus (*Alt1*) has been mapped to a small interval on the short arm of chromosome 6 (36). Whether *Alt1* in rye also has a cognate in wheat that participates in Al tolerance responses is unclear, as this region has not been implicated by cosegregation analysis in wheat.

Barley (*Hordeum vulgare*), a third member of the Triticeae tribe, also contains a major Al tolerance locus, *Alp*, on the long arm of chromosome 4 (106). Like *Alt3* in rye, the *Alp* locus is linked to markers useful for following *Alt_{BH}* in wheat (153) (see Figure 3A, see color insert). Unlike rye or wheat, barley is markedly sensitive to Al toxicity and demonstrates tolerance to only very low levels of stress (106). If *Alp*, *Alt3*, and *Alt_{BH}* are truly orthologous loci, one would expect that an analysis of the protein sequences should reveal a great deal about how this wide range of tolerance phenotypes is achieved.

Sorghum (*Sorghum bicolor* L. Moench) is a newcomer to the fields of Al tolerance genetics and molecular genetics. Sorghum is closely related to maize, possesses the second smallest genome among cultivated grasses, and exhibits a wide phenotypic range for biotic and abiotic stress tolerances, making this warm weather grass an attractive and new model experimental system (108). Recent investigations into the inheritance of Al tolerance indicate that like wheat, rye, and barley, sorghum exhibits a qualitative pattern of inheritance with a single locus explaining

most of the differences (98; J. Magalhaes & L. Kochian, unpublished results). Unlike the Triticeae, the *Alt_{SB}* locus in sorghum is neither located in the homeologous chromosomal location to *Alp*, *Alt3*, and *Alt_{BH}*, nor is it linked to the shared set of RFLPs and SSRs. The phenomenology of Al tolerance is also different in sorghum, relative to the Triticeae because the tolerance response is inducible and takes days to fully manifest (98). The combination of genetic and physiological data suggests that sorghum uses a different pathway to achieve Al tolerance than the mechanism characterized in wheat and its relatives uses.

Tolerance Loci with Quantitative Inheritance

Rice (*Oryza sativa*) is the best-characterized plant where Al tolerance is a quantitative trait (89, 112–114, 168). Four of the five studies with rice used mapping populations made between subspecies *indica* and *japonica* or between *indica* and a wild relative, *Oryza rufipogon* (89, 112, 114, 168). These mapping populations are all highly polymorphic and likely capture a great deal of the nucleotide diversity present in rice. Twenty-seven QTLs important for Al tolerance, as estimated by relative root growth, were identified in the five studies. Only one region on chromosome 1 was held in common between all five studies. Three other chromosomal regions (chromosomes 2, 3, and 9) were identified as important in three of the five mapping populations. Given the conservation of location for Al tolerance loci among the Triticeae, it is natural to wonder if an orthologous locus to *Alp/Alt3/Alt_{BH}* plays a similar role in rice. Rice chromosome 3 (linkage block 3C) is homeologous to Triticeae 4L (34); genetic markers linked to Al tolerance loci are shared between rice, wheat, and barley (Figure 3A) (112). However, the Al tolerance locus on rice 3 is not the principal one, instead the locus on rice 1 typically explains the largest variance. Surprisingly, the rice 1 QTL is in a region homeologous to the portion of sorghum linkage group G that contains *Alt_{SB}* (Figure 3B). Further work is necessary to evaluate whether orthologous loci are at work in both sorghum and rice, or if the apparent linkage is merely serendipitous.

Like wheat, maize (*Zea mays*) has been the subject of breeding programs seeking to increase Al tolerance or understand the basis for it (99). Some investigators have concluded that Al tolerance was a qualitative trait, although these studies utilized either small mapping populations (<100 F2 individuals) or nearly identical mapping parents (a tolerant inbred and a sensitive somaclonal variant) (130, 145). Most investigators have concluded that Al tolerance is a quantitative trait, based on the segregation analysis of large F2:3 populations or recombinant inbred lines (38, 99, 116). Only one QTL mapping study has been published to date; therein, five genomic regions were identified as important for Al tolerance (116). When we conducted an *in silico* comparative mapping analysis of these regions using the Gramene database (160), two of these regions in maize also contained Al tolerance loci in other grasses. The markers that flank a principal maize QTL region (bin 6.05) can be landed to rice chromosome 5, in the same vicinity of a

QTL identified by Nguyen and coworkers (113). The third most important QTL from the maize study (bin 8.07) falls within one of the two regions of the maize genome homeologous to rice chromosome 1 (linkage block 1B) and sorghum linkage group G, the chromosomal segment identified by six different studies (Figure 3B). As the physical map of maize improves, it will be possible to land the other three QTLs to their related chromosomal regions in rice in the future. There is a lot of exciting comparative mapping to do in the grasses to investigate the role of putative orthologs in Al tolerance.

Unlike many of the examples listed above, where investigators used varieties adapted to their own local conditions, workers in *Arabidopsis thaliana* often use precisely the same varieties. Two studies offer the opportunity to investigate how experimental conditions affect the outcome of the QTL analysis, as two different groups used the same mapping population (50, 66). In both studies, the principal QTL was found at the top of chromosome 1 and explained approximately 30% of the variance observed; however, the locations for the other putative QTLs were in complete disagreement between studies. This outcome is possibly due to the affect of the growth conditions on root growth in each study, as Kobayashi & Koyama (66) grew their seedlings in very low ionic strength hydroponic media at pH 5.0, whereas Hoekenga and coworkers (50) used higher ionic strength nutrient solution and Al concentration in a gelled growth media. Because phenotype is the result of both genetic and environmental effects, investigators must adopt standard environmental conditions/protocols in order to expect directly comparable results from their genetic analyses.

The study by Hoekenga et al. (50) demonstrates the importance of malate release in Al tolerance for the Landsberg *erecta* × Columbia RIL. A correlation analysis of Al tolerance and malate release indicated that almost all the variance in tolerance ($r^2 = 0.95$) was explained by malate exudation. The results from this single cross reinforce the study of Ryan et al. (137), where a single physiological mechanism, which may contain multiple, polymorphic components in a population, is responsible for the differences in Al tolerance.

Genomic Analysis of Al Tolerance

The physiological and genetic investigations on Al tolerance provide an excellent starting position for genomics-based inquiries. Based on the physiological characterizations of Al tolerance and toxicity, it is clear that OA release and internal detoxification are key players in Al tolerance pathways. Understanding these pathways from a mechanistic standpoint improves candidate gene selection during map-based cloning. In several systems, Al tolerance may be an inducible process (see, for example 78, 98), and this indicates that profiling the changing patterns of gene or protein expression may permit the identification of factors important to tolerance. This section summarizes progress to date in identifying genes and proteins important for Al tolerance and attempts to confirm the importance of these candidate genes using transgenic plants or yeast.

GENE EXPRESSION PROFILING Experiments in wheat have also led efforts to profile changes in gene expression during Al stress responses. Several groups have investigated genes expressed in Al-treated wheat roots (42, 105, 132, 152). Genes identified from these experiments were largely identified as responsive to other abiotic or biotic stresses and most were expressed to the same degree in both sensitive and tolerant genotypes (see 105 for a list of Al-inducible genes). With few exceptions, the relevance of Al-inducible genes to Al tolerance processes has yet to be demonstrated.

Delhaize and coworkers identified genes from wheat in a more functional manner, via screening of *S. cerevisiae* transformed with plant genes for changes in Al tolerance (19). This strategy identified six unique clones, one of which encoded a phosphatidylserine synthase, which is involved in phospholipid metabolism. Although overexpression of the gene conferred Al tolerance to yeast, the same was not true in *Arabidopsis*, where instead spontaneous necrotic lesions and dwarfing occurred (19). This was likely related to the disturbance in the balance between phosphatidylserine and phosphatidylglycerol, where the latter is a major phospholipid in leaves and an essential component of chloroplast lamellae. The relationship between Al toxicity and lipid peroxidation is well known (for example 170, 171); it is possible that phosphatidylserine is less susceptible to peroxidation, such that an increased abundance of this phospholipid might protect membranes in yeast.

Subtractive cDNA libraries have also been constructed from *Arabidopsis*, wheat, tobacco suspension cells, rye, and sugarcane (*Saccharum* spp.) to identify Al-inducible genes (26, 29, 30, 105, 131, 162). The best demonstration for relevance to Al tolerance for the genes identified comes from the construction of transgenic plants, especially in *Arabidopsis*, where transformation is easy and efficient (Table 2); transgenics were subsequently challenged with Al and their tolerance was evaluated (27, 28, 149). Ezaki and coworkers had only modest success in increasing the Al tolerance of *Arabidopsis* with their transgenes, with gains of 50% or less (from 60% inhibition of root growth in untransformed to 40% inhibition in transgenics) (27). Anionic peroxidase from tobacco produced the largest and most consistent increase in Al tolerance in these experiments, presumably by increasing the capacity of the plant to cope with ROS. Basu and coworkers produced a similar result working in *Brassica napus*, where they overexpressed a manganese superoxide dismutase and observed a modest gain in Al tolerance (9). Two other examples of transgenic expression and evaluation of the candidate Al tolerance genes *ALMT1* from wheat and *Arabidopsis WAK1* are discussed above (142, 149).

One benefit of genomic research is the generation of large collections of gene expression data, which can then be accessed by all members of the community. An example is the dataset generated by the *Arabidopsis* Functional Genomics Consortium (AFGC); all of the data generated from the microarray hybridizations are searchable via The *Arabidopsis* Information Resource (TAIR) website (31) (<http://www.Arabidopsis.org>). Two microarray hybridizations were performed at the AFGC using RNA isolated from Al-treated *Arabidopsis* (50). As these data

TABLE 2 Genes expressed in transgenic plants to increase Al tolerance. The outcome of each transformation experiment is expressed as no change in Al tolerance (=), improvement in Al tolerance (+), or a detrimental effect to plant growth (−)

Gene	Origin	Recipient	Outcome	References
ALR1 (Mg transporter)	<i>Saccharomyces cerevisiae</i>	<i>S. cerevisiae</i>	+	(96)
AtBCB (blue copper binding protein)	<i>Arabidopsis thaliana</i>	<i>A. thaliana</i>	+	(27)
AtBPI (protease inhibitor)	<i>A. thaliana</i>	<i>A. thaliana</i>	=	(27)
AtPOX (peroxidase)	<i>A. thaliana</i>	<i>A. thaliana</i>	=	(27)
Citrate synthase	<i>Pseudomonas aeruginosa</i>	<i>Nicotiana tabacum</i> , <i>Carica papaya</i>	+	(17)
Citrate synthase	<i>A. thaliana</i>	<i>Daucus carota</i>	+	(72)
Citrate synthase	<i>P. aeruginosa</i>	<i>N. tabacum</i>	=	(20)
Citrate synthase	<i>A. thaliana</i>	<i>Brassica napus</i>	+	(5)
HSP150 (heat shock protein)	<i>S. cerevisiae</i>	<i>A. thaliana</i>	=	(27)
Malate dehydrogenase	<i>Medicago sativa</i>	<i>M. sativa</i>	+	(156)
Mn SOD	<i>Triticum aestivum</i>	<i>B. napus</i>	+	(9)
NtGDI (GDP dissociation inhibitor)	<i>N. tabacum</i>	<i>A. thaliana</i>	+	(27)
NtPOX (anionic peroxidase)	<i>N. tabacum</i>	<i>A. thaliana</i>	+	(27)
parA	<i>N. tabacum</i>	<i>A. thaliana</i>	=	(27)
parB (Glutathione S-transferase)	<i>N. tabacum</i>	<i>A. thaliana</i>	+	(27)
PEP carboxylase	<i>M. sativa</i>	<i>M. sativa</i>	=	(156)
Phosphatidylserine synthase	<i>T. aestivum</i>	<i>S. cerevisiae</i> <i>thaliana</i> <i>N. tabacum</i>	+	(19)
wak1 (wall-associated kinase)	<i>A. thaliana</i>	<i>A. thaliana</i>	+	(149)
Wali5 (protease inhibitor)	<i>T. aestivum</i>	<i>A. thaliana</i>	=	(27)

were uploaded to the public website shortly after the microarray experiments were complete, Schultz and coworkers identified that the Al treatment induced two members and repressed two others of the 19-member arabinogalactan protein (AGP) family. AGPs are cell wall-localized proteins that are important for growth and development, but are poorly characterized (143). AGP2 is Al inducible, with maximal expression after 8 h of stress; it is unknown what role AGP2 may play in

an Al stress response, although the specificity of the response by this member of the AGP gene family is suggestive (143).

PROTEIN EXPRESSION AND ACTIVITY PROFILING The effect of Al stress on root protein profiles has been a topic of investigation longer than gene expression profiling experiments (see 67). Since the earliest surveys of root tip proteomes, workers have concentrated their efforts in three areas: identifying potential tolerance proteins, identifying toxicity sites, and attempting to increase tolerance directly by modifying existing pathways.

Taylor and coworkers identified an Al-inducible 51-kDa protein, which they found in microsomal preparations from an Al-tolerant cultivar of wheat (8). This 51-kDa protein was a pair of proteins; both proteins were identified as ATP synthase subunits, one vacuolar and one mitochondrial (43). The V-ATPase β subunit was also an Al-inducible gene, whereas the F_1F_0 -ATPase α subunit was constitutively expressed (43). It is not clear how either ATPase could be involved in Al tolerance, although the authors speculate that the vacuolar H^+ -ATPase might be involved in maintaining cytoplasmic pH homeostasis if this is disturbed during Al toxicity due to Al inhibition of the plasma membrane H^+ pump.

MacDiarmid & Gardner (95) demonstrated that an additional facet of Al toxicity might come from magnesium uptake inhibition. This was initially identified in Al-sensitive mutants in *S. cerevisiae*, where the lesion was localized to a plasma membrane Mg^{2+} transporter. This observation was extended using yeast strains that overexpressed the *ALR1* and *ALR2* Mg^{2+} transporters; these recombinant strains had increased Al tolerance relative to the untransformed parent strain (96). These findings became more relevant to plant research when a family of Mg^{2+} transporters was characterized from *Arabidopsis* (AtMGT); AtMGT1 and AtMGT10 can functionally complement yeast strains deficient in Mg^{2+} transporters (77). Both of these proteins are inhibited by Al, as are their yeast homologs, suggesting that Al toxicity can affect Mg^{2+} uptake in *Arabidopsis* (77).

Several attempts to increase Al tolerance via transgenes are described above. In most cases, relatively anonymous, Al-inducible genes were selected as transformation targets (9, 27, 28, 149). However, there have been attempts to increase Al tolerance by increasing OA release in response to Al stress (Table 2). These experiments revolved around the premise that increasing the abundance of enzymes related to OA synthesis, as this is the only part of this mechanism where genes have been cloned, might increase the efflux of ligands from roots (5, 17, 20, 22, 72, 156). The results from these studies have been mixed, where some investigators reported modest increases in tolerance (17, 72), whereas others using the same constructs saw no improvement (20). The results of these experiments, in part, suggest that ligand availability is not the limiting step in Al-activated OA release, which is supported by the physiological investigations of Al-activated OA exudation described above. In the future, transgenic approaches may require the coexpression of a specific OA transporter along with the appropriate biosynthetic enzymes in order to optimize the enhancement of Al tolerance.

PHOSPHOROUS EFFICIENCY: TOLERANCE TO P DEFICIENCY STRESS

In almost all soils, P is often the most limiting mineral nutrient. Its bioavailability is low due to binding to soil mineral surfaces and fixation into organic forms; available phosphate (Pi) in the soil solution is commonly 1–2 μM (13). Pi availability is particularly limiting on the highly weathered, acid soils of the tropics and subtropics, due to its fixation with Al and Fe oxides on the surface of clay minerals. Hence, P availability is a major factor limiting crop production on acid soils (141). Because of the low availability of this essential mineral nutrient, plants have evolved numerous adaptive mechanisms to acquire Pi from the soil. These responses include increases in root proliferation including root branching, increased root hairs, and association with vesicular-arbuscular mycorrhizae aimed at exploring a more extensive volume of soil for P, relocation of C resources to the root, root-mediated changes in rhizosphere chemistry aimed at increasing P availability, and upregulation of Pi transporters (128).

There is significant genetically based inter- and intraspecific variation in the ability of plants to tolerate P deficiency stress, a trait termed P efficiency (82, 83). P efficiency can be based on the superior ability to acquire P from the soil through alterations in root morphology or architecture, exudation of P mobilizing compounds, and alterations in plasma membrane Pi transporters. Additionally, it could involve enhanced P use efficiency through lower cellular P requirements or more efficient remobilization of P within the plant (172). Research into mechanisms of P efficiency is in its early stages; hence, here we address some important plant adaptations to P deficiency as well as how they might be involved in P efficiency.

Root Exudates and P Mobilization from the Soil

There are several published reports indicating that P deficiency can trigger the exudation of malate and citrate from the root (see 135 and the references therein). These OA anions can desorb Pi from mineral surfaces, solubilizing it from associations with Al, Fe, and Ca oxides and hydroxides via metal complexation. The most dramatic example of this is the induction of dense clusters of lateral or proteoid roots in members of the Proteaceae family (e.g., *Lupinus albus*) (111) under P deficiency. In synchrony with this, P deficiency also triggers changes in cellular OA metabolism and transport such that high rates of citrate and malate exudation from these proteoid roots occur, helping to solubilize soil P for uptake (see, for example, 56, 57). The loss of C due to OA efflux from proteoid roots can exceed 20% of the total plant dry weight in *L. albus* (24).

P deficiency conditions also induce the root exudation of phosphatases and RNAases in many plant species (see, for example, 117, 161). Because a significant fraction of soil P can be fixed in organic compounds, the activity of these enzymes could play roles in catalyzing the hydrolysis of organic P for uptake by roots. However, the role of phosphatases in liberating P from organic sources in the soil

is still debatable, as there have been reports of both positive (6, 47) and negative (103) correlations between root phosphatase exudation and P acquisition, as well as a report showing no significant difference in root surface phosphatases between white clover genotypes contrasting in P efficiency (54).

Root Morphology and Architecture

Because diffusion of Pi in the soil is slow, P acquisition strategies aimed at expanding the root surface area in contact with the soil may be an effective way to increase P efficiency. There is evidence in support of this strategy. (a) During P deficiency, there is a significant increase in the root:shoot ratio and concomitant reallocation of C resources to support this (82). (b) Phosphorus deficiency causes significant changes in the morphology, geometry, and architecture of the root system (80, 82, 165). These changes are characterized by an increase in root branching, and a decrease in the root diameter, thus increasing the amount of absorptive surface area of the root relative to the volume. Both of these alterations allow the root system to explore a larger volume of soil (12). (c) The enhanced production, elongation, and proliferation of root hairs (10) increase the root surface area for Pi acquisition (10, 11, 93). (d) As described above, P deficiency leads to the development of cluster or proteoid roots with significant OA exudation in certain plant species (for reviews, see 73, 111). (e) The symbiosis of vesicular-arbuscular mycorrhizae (VAM) with plant roots of almost all higher plant species clearly enhances the uptake of diffusion-limited mineral nutrients from the soil, in particular P and Zn (see 44, 69 for reviews). Because these associations are widespread and found in almost all crop plant species, it is debatable whether differences in plant P efficiency involve VAM.

Pi Homeostasis and Transport

P deficiency induces a significant enhancement in net Pi uptake into plant roots. This is due both to a stimulation of unidirectional Pi influx via increased expression of Pi transporters, and a reduction in P efflux. At the whole-plant level, P deficiency causes a reallocation and translocation of P from older plant tissues to growing organs. At the cellular level, P homeostasis involves regulating exchange between various intracellular pools to maintain cytoplasmic Pi levels within a narrow concentration range (approximately 5–10 mM) (128). Thus, during the onset of P deficiency, Pi vacuolar pools are preferentially depleted whereas cytoplasmic P levels remain fairly constant.

The concentration-dependent kinetics of root Pi uptake indicates that a dual uptake model is functioning, involving a high-affinity transporter active at low (μM) concentration and a lower affinity transporter active at higher Pi concentrations. Given the low (μM) concentration of available Pi in natural soil solutions, the high-affinity transporter is the primary mechanism mediating Pi uptake. Based on the thermodynamic considerations for Pi uptake as well as the pH dependency and observations of a net alkalization of the rhizosphere during Pi uptake, the

general consensus is that root Pi uptake is mediated via a thermodynamically active H^+ -Pi cotransporter driven by the plasma membrane H^+ -ATPase (128). This is supported by molecular studies that have led to the isolation of Pi transporter genes from numerous plant species; analysis of the deduced amino acid sequence for these transporters indicates that they are a subfamily of the Major Facilitator Superfamily of transporters that include most of the plant H^+ -coupled cotransporters that mediate the active uptake of sugars, amino acids, and inorganic anions (128).

Genetic Analysis of P Efficiency

Scientists are using two different genetic approaches to study plant P deficiency responses to provide information on P efficiency: identifying and characterizing *Arabidopsis* mutants with altered P nutrition, and analyzing the qualitative or quantitative inheritance in populations segregating for P efficiency. A number of *Arabidopsis* P nutrition mutants have been identified, including two P transport mutants (*pho1* and *pho2*), a mutant lacking a root acid phosphatase (*pup1*; 158), and several putative P deficiency response mutants (*pho3*, *psr1*). *Pho1* was the first *Arabidopsis* P nutrition mutant identified, and it has very low levels of shoot P, whereas root P concentrations are normal (127). Physiological characterization of this mutant indicated it was defective in root-to-shoot P transport, and it was proposed that the mutation led to defective loading of Pi into the xylem. Subsequently, the *pho1* gene was isolated via a map-based cloning strategy and was identified as a member of a novel class of membrane proteins that do not share homology with P transporters from plants or other organisms (41). Expression analysis indicated *pho1* is expressed in the root stele, which is consistent with its possible function in loading P into the xylem. The *pho2* mutant accumulates excessive P in its shoots, and a detailed physiological characterization of this mutant showed that its phenotype could be due to a defect in Pi transport between shoots and roots (25). The *pho3* and *psr1* mutants were isolated based on screens for either impaired root acid phosphatase exudation and activity (*pho3*) (174) or impaired ability to acquire P when DNA or RNA was the only source for roots (*psr1*) (15). In both cases, the authors suggest these mutations are defects in the plant's ability to sense and/or respond to changes in plant P status, and thus should be useful in beginning to elucidate the P signaling pathways in higher plants. This field is in its infancy, and the possible role in P efficiency for the genes underlying these mutations remains to be determined.

Several studies in *Arabidopsis* have looked at the genetic variation in P efficiency and have set the stage for subsequent inheritance studies. Narang et al. (110) screened 36 different *Arabidopsis* accessions for P efficiency based on their ability to grow on a relatively insoluble P source (hydroxylapatite). They focused on the five accessions that represented the extremes in P efficiency and found different specific root traits associated with the three most P-efficient ecotypes (*C24*, *Co*, *Cal*) that were superior to the inefficient accessions (*Col-0*, *Te*), including changes in root morphology, OA release, rhizosphere acidification, root hair length and

density, and root penetration. A subsequent analysis of the F1 hybrid from the cross of *C24* (P efficient) with inefficient *Col-0* showed that the F1 exhibited superior P efficiency to both parents (109). Physiological analysis of the F1 plants indicated they had accumulated superior alleles for P efficiency from both parents.

Two groups have conducted QTL mapping for P efficiency by using different mapping populations in rice (115, 167). Both studies identified a major QTL for P efficiency in a similar region of chromosome 12, and a minor QTL on chromosome 6. Wissuwa et al.'s (167) study also identified minor QTLs on chromosomes 2 and 10, whereas Ni et al. (115) found minor QTLs on chromosomes 1 and 9. Although one might assume that the ideal P efficiency strategy should combine high-P uptake and efficient internal P utilization, these two studies suggest that rice P efficiency is mainly due to genotypic differences in root P uptake. Differences in internal P use efficiency were minimal. Further evidence in support of the major QTL for P efficiency on chromosome 12 as well as the minor one on chromosome 6 comes from a subsequent study by Wissuwa & Ae (166) that uses near-isogenic lines (NILs) developed for both of these QTL. Under low-P growth conditions, the NILs for the major QTL on chromosome 12 had three- to fourfold higher P uptake compared with the P-inefficient parent (Nipponbare), whereas the NIL representing the minor QTL showed 60% to 90% higher P uptake. These studies suggest that P efficiency is a complex trait, and although a major QTL was identified, it will not be known whether this QTL is due to the action of one gene or several genes until the genes underlying the QTL have been cloned.

SUMMARY

Over the past decade, there has been significant progress in our understanding of the mechanistic basis for crop acid soil tolerance, as well as progress toward the goal of developing crops better suited for cultivation on acid soils. Several physiological aspects of Al-activated root OA exudation in numerous plant species have been uncovered, and this research has raised as many questions for subsequent inquiry as it has answered. We are just beginning to understand a second Al tolerance mechanism involving internal detoxification of Al with OA ligands and the sequestration of the Al OA complexes in the vacuole. Researchers in the field are also becoming increasingly aware that acid soil tolerance is not just Al tolerance, but also involves the enhanced ability to acquire P and other limiting nutrients, as well as tolerances to other toxic metals. We expect that major advances in the physiological and molecular basis for P efficiency and Mn tolerance will occur in the next decade. Scientists in this field are on the brink of identifying a number of genes that are key players in Al tolerance; these discoveries will open up new avenues of examination into the molecular and genetic basis for Al tolerance, as well as provide new molecular resources for further improvements in crop Al tolerance via both molecular-assisted plant breeding and biotechnology.

ACKNOWLEDGMENTS

The authors acknowledge the USDA-CSREES National Research Initiative Competitive Grants Program (USDA-NRI Grants #00–35,100-9280, #01–35,301-10,647, and #02–35,100-12,058), the McKnight Foundation Collaborative Crop Research Program, and the USDA Agricultural Research Service for financial support for research on acid soil tolerance.

The *Annual Review of Plant Biology* is online at <http://plant.annualreviews.org>

LITERATURE CITED

1. Akeson M, Munns DN. 1990. Uptake of aluminum into root cytoplasm: predicted rates for important solution complexes. *J. Plant Nutr.* 13:467–84
2. Anderson CM, Wagner TA, Perret M, He ZH, He D, Kohorn BD. 2001. WAKs: cell wall-associated kinases linking the cytoplasm to the extracellular matrix. *Plant Mol. Biol.* 47:197–206
3. Anderson MP, Gregory RJ, Thompson S, Souza DW, Paul S, et al. 1991. Demonstration that CFTR is a chloride channel by alteration of its anion selectivity. *Science* 253:202–5
4. Aniol A, Gustafson JP. 1984. Chromosome location of genes controlling aluminum tolerance in wheat, rye and triticale. *Can. J. Genet. Cytol.* 26:701–5
5. Anoop VM, Basu U, McCammon MT, McAlister-Henn L, Taylor GJ. 2003. Modulation of citrate metabolism alters aluminum tolerance in yeast and transgenic canola overexpressing a mitochondrial citrate synthase. *Plant Physiol.* 132:2205–17
6. Asmar F, Gahoonia T, Nielsen N. 1995. Barley genotypes differ in activity of soluble extracellular phosphatase and depletion of organic phosphorus in the rhizosphere soil. *Plant Soil* 172:117–22
7. Barcelo J, Poschenrieder C. 2002. Fast root growth responses, root exudates, and internal detoxification as clues to the mechanisms of aluminium toxicity and resistance: A review. *Environ. Exp. Bot.* 48:75–92
8. Basu A, Basu U, Taylor GJ. 1994. Induction of microsomal membrane proteins in roots of an aluminum-resistant cultivar of *triticum aestivum* L. under conditions of aluminum stress. *Plant Physiol.* 104:1007–13
9. Basu U, Good AG, Taylor GJ. 2001. Transgenic *Brassica napus* plants overexpressing aluminum-induced mitochondrial manganese superoxide dismutase cDNA are resistant to aluminum. *Plant Cell Environ.* 24:1269–78
10. Bates T, Lynch J. 1996. Stimulation of root hair elongation in *Arabidopsis thaliana* by low phosphorus availability. *Plant Cell Environ.* 19:529–38
11. Bates T, Lynch J. 2000. The efficiency of *Arabidopsis thaliana* (Brassicaceae) root hairs in phosphorus acquisition. *Am. J. Bot.* 87:964–70
12. Bates T, Lynch J. 2001. Root hairs confer a competitive advantage under low phosphorus availability. *Plant Soil* 236:243–50
13. Bielecki RL. 1973. Phosphate pools, phosphate transport, and phosphate availability. *Annu. Rev. Plant Physiol.* 24:225–52
14. Blancaflor EB, Jones DL, Gilroy S. 1998. Alterations in the cytoskeleton accompany aluminum-induced growth inhibition and morphological changes in primary roots of maize. *Plant Physiol.* 118:159–72

15. Chen D, Delatorre C, Bakker A, Abel S. 2000. Conditional identification of phosphate-starvation-response mutants in *Arabidopsis thaliana*. *Planta* 211:13–22
16. Degenhardt J, Larsen PB, Howell SH, Kochian LV. 1998. Aluminum resistance in the *Arabidopsis* mutant alr-104 is caused by an aluminum-induced increase in rhizosphere pH. *Plant Physiol.* 117:19–27
17. de la Fuente JM, Ramirez-Rodriguez V, Cabrera-Ponce JL, Herrera-Estrella L. 1997. Aluminum tolerance in transgenic plants by alteration of citrate synthesis. *Science* 276:1566–8
18. Delhaize E, Craig S, Beaton CD, Bennet RJ, Jagadish VC, Randall PJ. 1993. Aluminum tolerance in wheat (*Triticum aestivum* L.): I. Uptake and distribution of aluminum in root apices. *Plant Physiol.* 103:685–93
19. Delhaize E, Hebb DM, Richards KD, Lin JM, Ryan PR, Gardner RC. 1999. Cloning and expression of a wheat (*Triticum aestivum* L.) phosphatidylserine synthase cDNA. Overexpression in plants alters the composition of phospholipids. *J Biol. Chem.* 274:7082–88
20. Delhaize E, Hebb DM, Ryan PR. 2001. Expression of a *Pseudomonas aeruginosa* citrate synthase gene in tobacco is not associated with either enhanced citrate accumulation or efflux. *Plant Physiol.* 125:2059–67
21. Delhaize E, Kataoka T, Hebb DM, White RG, Ryan PR. 2003. Genes encoding proteins of the cation diffusion facilitator family that confer manganese tolerance. *Plant Cell* 15:1131–42
22. Delhaize E, Ryan PR, Hocking PJ, Richardson AE. 2003. Effects of altered citrate synthase and isocitrate dehydrogenase expression on internal citrate concentrations and citrate efflux from tobacco (*Nicotiana tabacum* L.) roots. *Plant Soil* 248:137–44
23. Delhaize E, Ryan PR, Randall PJ. 1993. Aluminum tolerance in wheat (*Triticum aestivum* L.): II. Aluminum-stimulated excretion of malic acid from root apices. *Plant Physiol.* 103:695–702
24. Dinkelaker B, Römheld V, Marschner H. 1989. Citric acid excretion and precipitation of calcium citrate in the rhizosphere of white lupin (*Lupinus albus* L.). *Plant Cell Environ.* 12:285–92
25. Dong B, Rengel Z, Delhaize E. 1998. Uptake and translocation of phosphate by *pho2* mutant and wild-type seedlings of *Arabidopsis thaliana*. *Planta* 205:251–56
26. Drummond RD, Guimaraes CT, Felix J, Ninamango CFE, Carneiro NP, et al. 2001. Prospecting sugarcane genes involved in aluminum tolerance. *Genet. Mol. Biol.* 24:221–30
27. Ezaki B, Gardner RC, Ezaki Y, Matsumoto H. 2000. Expression of aluminum-induced genes in transgenic *Arabidopsis* plants can ameliorate aluminum stress and/or oxidative stress. *Plant Physiol.* 122:657–65
28. Ezaki B, Katsuhara M, Kawamura M, Matsumoto H. 2001. Different mechanisms of four aluminum (Al)-resistant transgenes for Al toxicity in *Arabidopsis*. *Plant Physiol.* 127:918–27
29. Ezaki B, Tsugita S, Matsumoto H. 1996. Expression of a moderately anionic peroxidases is induced by aluminum treatment in tobacco cells: possible involvement of peroxidase isozymes in aluminum ion stress. *Physiol. Plant* 96:21–28
30. Ezaki B, Yamamoto Y, Matsumoto H. 1995. Cloning and sequencing of the cDNAs induced by aluminum treatment and Pi starvation in cultured tobacco cells. *Physiol. Plant.* 93:11–18
31. Finkelstein D, Ewing R, Gollub J, Sterky F, Cherry JM, Somerville S. 2002. Microarray data quality analysis: lessons from the AFGC project. *Arabidopsis Functional Genomics Consortium. Plant Mol. Biol.* 48:119–31
32. Foy C. 1984. Physiological effects of

- hydrogen, aluminum, and manganese toxicities in acid soils. In *Soil Acidity and Liming*, ed. F Adams, pp. 57–97. Madison, WI: Agron. Soc. Am.
33. Foy C, Flemming A, Schwartz J. 1973. Opposite aluminum and manganese tolerances in two wheat varieties. *Agron. J.* 65:123–26
34. Gale MD, Devos KM. 1998. Comparative genetics in the grasses. *Proc. Natl. Acad. Sci. USA* 95:1971–74
35. Gallego FJ, Benito C. 1997. Genetic control of aluminium tolerance in rye (*Secale cereale* L.). *Theor. Appl. Genet.* 95:393–99
36. Gallego FJ, Calles B, Benito C. 1998. Molecular markers linked to the aluminium tolerance gene *alt1* in rye (*Secale cereale* L.). *Theor. Appl. Genet.* 97:1104–9
37. Garvin DF, Carver BF. 2003. Role of the genotype in tolerance to acidity and aluminum toxicity. In *Handbook of Soil Acidity*, ed. Z Rengel, pp. 387–406. New York: Marcel Dekker
38. Giaveno GD, Miranda FJB, Furlani PR. 2001. Inheritance of aluminum tolerance in maize (*Zea mays* L.). *J. Gen. Breed.* 55:51–56
39. Gonzalez A, Steffen KL, Lynch JP. 1998. Light and excess manganese. Implications for oxidative stress in common bean. *Plant Physiol.* 118:493–504
40. Grabski S, Schindler M. 1995. Aluminum induces rigor within the actin network of soybean cells. *Plant Physiol.* 108:897–901
41. Hamburger D, Rezzonico E, MacDonald-Comber Petetot J, Somerville C, Poirier Y. 2002. Identification and characterization of the Arabidopsis PHO1 gene involved in phosphate loading to the xylem. *Plant Cell* 14:889–902
42. Hamel F, Breton C, Houde M. 1998. Isolation and characterization of wheat aluminum-regulated genes: possible involvement of aluminum as a pathogenesis response elicitor. *Planta* 205:531–38
43. Hamilton CA, Good AG, Taylor GJ. 2001. Induction of vacuolar ATPase and mitochondrial ATP synthase by aluminum in an aluminum-resistant cultivar of wheat. *Plant Physiol.* 125:2068–77
44. Harrison MJ. 1999. Molecular and cellular aspects of the arbuscular mycorrhizal symbiosis. *Annu. Rev. Plant Physiol. Plant Mol. Biol.* 50:361–89
45. Hayes JE, Ma JF. 2003. Al-induced efflux of organic acid anions is poorly associated with internal organic acid metabolism in triticale roots. *J. Exp. Bot.* 54:1753–59
46. Hede AR, Skovmand B, Lopez-Cesati J. 2001. Acid soils and aluminum toxicity. In *Application of Physiology in Wheat Breeding*, ed. MP Reynolds, JI Ortiz-Monasterio, A McNab, pp. 172–82. Mexico: CIMMYT
47. Helal HM. 1990. Varietal differences in root phosphatase activity as related to the utilization of organic phosphates. *Plant Soil* 123:161–64
48. Hirschi KD, Korenkov VD, Wilganowski NL, Wagner GJ. 2000. Expression of Arabidopsis CAX2 in tobacco. Altered metal accumulation and increased manganese tolerance. *Plant Physiol.* 124:125–33
49. Hoekenga O, Mason P, Shaff J, Buckler E, Kochian L. 2003. Identification and characterization of Al tolerance genes in the intermated B73 × Mo17 population of maize by QTL mapping. *Plant Physiol.*, p. 53 (Abstr.)
50. Hoekenga OA, Vision TJ, Shaff JE, Monforte AJ, Lee GP, et al. 2003. Identification and characterization of aluminum tolerance loci in Arabidopsis (*Landsberg erecta* × *Columbia*) by quantitative trait locus mapping. A physiologically simple but genetically complex trait. *Plant Physiol.* 132:936–48
51. Horst W, Asher C, Cakmak I, Szulkiewicz P, Wissemeier AH. 1992. Short-term responses of soybean roots to aluminium. *J. Plant Physiol.* 140:174–78
52. Horst W, Fecht M, Naumann A, Wissemeier A, Maier P. 1999. Physiology of

- manganese toxicity and tolerance in *Vigna unguiculata* (L.) Walp. *J. Plant Nutr.* 162:263–74
53. Huang JW, Pellet DM, Papernik LA, Kochian LV. 1996. Aluminum interactions with voltage-dependent calcium transport in plasma membrane vesicles isolated from roots of aluminum-sensitive and -resistant wheat cultivars. *Plant Physiol.* 110:561–69
54. Hunter D, Leung D. 1999. Comparison of acid phosphatases in two genotypes of white clover with different responses to applied phosphate. *J. Plant Nutr.* 22:679–92
55. Ishikawa S, Wagatsuma T, Sasaki R, Ofey-Manu P. 2000. Comparison of the amount of citric and malic acids in Al media of seven plant species and two cultivars each in five plant species. *Soil Sci. Plant Nutr.* 46:751–58
56. Johnson JF, Allan DL, Vance CP. 1994. Phosphorus stress-induced proteoid roots show altered metabolism in *Lupinus albus*. *Plant Physiol.* 104:657–65
57. Johnson JF, Vance CP, Allan DL. 1996. Phosphorus deficiency in *Lupinus albus*. Altered lateral root development and enhanced expression of phosphoenolpyruvate carboxylase. *Plant Physiol.* 112:31–41
58. Jones DL, Gilroy S, Larsen PB, Howell SH, Kochian LV. 1998. Effect of aluminum on cytoplasmic Ca^{2+} homeostasis in root hairs of *Arabidopsis thaliana* (L.). *Planta* 206:378–87
59. Jones DL, Kochian LV. 1995. Aluminum inhibition of the inositol 1,4,5-triphosphate signal transduction pathway in wheat roots: a role in aluminum toxicity? *Plant Cell* 7:1913–22
60. Jones DL, Kochian LV, Gilroy S. 1998. Aluminum induces a decrease in cytosolic calcium concentration in BY-2 tobacco cell cultures. *Plant Physiol.* 116:81–89
61. Jorge R, Arruda P. 1997. Aluminum-induced organic acid exudation by roots of an aluminum-tolerant tropical maize. *Phytochemistry* 45:675–81
62. Kidd PS, Llugany M, Poschenrieder C, Gunse B, Barcelo J. 2001. The role of root exudates in aluminium resistance and silicon-induced amelioration of aluminium toxicity in three varieties of maize (*Zea mays* L.). *J. Exp. Bot.* 52:1339–52
63. Kinraide T, Parker D. 1989. Assessing the phytotoxicity of mononuclear hydroxy-aluminum. *Plant Cell Environ.* 12:478–87
64. Kinraide T, Parker D. 1990. Apparent phytotoxicity of mononuclear hydroxy-aluminum to four dicotyledonous species. *Physiol. Plant.* 79:283–88
65. Kinraide TB. 1991. Identity of the rhizotoxic aluminum species. *Plant Soil* 134:167–78
66. Kobayashi Y, Koyama H. 2002. QTL analysis of Al tolerance in recombinant inbred lines of *Arabidopsis thaliana*. *Plant Cell Physiol.* 43:1526–33
67. Kochian LV. 1995. Cellular mechanisms of aluminum toxicity and resistance in plants. *Annu. Rev. Plant Physiol. Plant Mol. Biol.* 46:237–60
68. Kochian LV, Jones DL. 1997. Aluminum toxicity and resistance in plants. In *Research Issues in Aluminum Toxicity*, ed. R Yokel, MS Golub, pp. 69–90. Washington, DC: Taylor & Francis
69. Koide RT, Schreiner RP. 1992. Regulation of the vesicular-arbuscular mycorrhizal symbiosis. *Annu. Rev. Plant Physiol.* 43:557–81
70. Kollmeier M, Dietrich P, Bauer CS, Horst WJ, Hedrich R. 2001. Aluminum activates a citrate-permeable anion channel in the aluminum-sensitive zone of the maize root apex. A comparison between an aluminum-sensitive and an aluminum-resistant cultivar. *Plant Physiol.* 126:397–410
71. Koyama H, Kawamura A, Kihara T, Hara T, Takita E, Shibata D. 2000. Overexpression of mitochondrial citrate synthase in *Arabidopsis thaliana* improved growth

- on a phosphorus-limited soil. *Plant Cell Physiol.* 41:1030–37
72. Koyama H, Takita E, Kawamura A, Hara T, Shibata D. 1999. Over expression of mitochondrial citrate synthase gene improves the growth of carrot cells in Al-phosphate medium. *Plant Cell Physiol.* 40:482–88
73. Lamont B. 2003. Structure, ecology and physiology of root clusters—a review. *Plant Soil* 248:1–19
74. Larsen PB, Degenhardt J, Tai CY, Stenzler LM, Howell SH, Kochian LV. 1998. Aluminum-resistant Arabidopsis mutants that exhibit altered patterns of aluminum accumulation and organic acid release from roots. *Plant Physiol.* 117:9–18
75. Lazof DB, Goldsmith JG, Rufty TW, Linton RW. 1996. The early entry of Al into cells of intact soybean roots. A comparison of three developmental root regions using secondary ion mass spectrometry imaging. *Plant Physiol.* 112:1289–300
76. Leonhardt N, Vavasseur A, Forestier C. 1999. ATP binding cassette modulators control abscisic acid-regulated slow anion channels in guard cells. *Plant Cell* 11:1141–52
77. Li L, Tutone AF, Drummond RS, Gardner RC, Luan S. 2001. A novel family of magnesium transport genes in Arabidopsis. *Plant Cell* 13:2761–75
78. Li XF, Ma JF, Matsumoto H. 2000. Pattern of aluminum-induced secretion of organic acids differs between rye and wheat. *Plant Physiol.* 123:1537–44
79. Liang F, Cunningham KW, Harper JF, Sze H. 1997. ECA1 complements yeast mutants defective in Ca^{2+} pumps and encodes an endoplasmic reticulum-type Ca^{2+} -ATPase in *Arabidopsis thaliana*. *Proc. Natl. Acad. Sci. USA* 94:8579–84
80. Lopez-Bucio J, Hernandez-Abreu E, Sanchez-Calderon L, Nieto-Jacobo MF, Simpson J, Herrera-Estrella L. 2002. Phosphate availability alters architecture and causes changes in hormone sensitivity in the Arabidopsis root system. *Plant Physiol.* 129:244–56
81. Luo M, Dvorak J. 1996. Molecular mapping of an aluminum tolerance locus on chromosome 4D of Chinese Spring wheat. *Euphytica* 91:31–35
82. Lynch J. 1995. Root architecture and plant productivity. *Plant Physiol.* 109:7–13
83. Lynch J. 1998. The role of nutrient efficient crops in modern agriculture. *J. Crop Prod.* 1:241–64
84. Ma JF, Furukawa J. 2003. Recent progress in the research of external Al detoxification in higher plants: a minireview. *J. Inorg. Biochem.* 97:46–51
85. Ma JF, Hiradate S. 2000. Form of aluminium for uptake and translocation in buckwheat (*Fagopyrum esculentum* Moench). *Planta* 211:355–60
86. Ma JF, Hiradate S, Matsumoto H. 1998. High aluminum resistance in buckwheat: II. Oxalic acid detoxifies aluminum internally. *Plant Physiol.* 117:753–59
87. Ma JF, Hiradate S, Nomoto K, Iwashita T, Matsumoto H. 1997. Internal detoxification mechanism of Al in hydrangea: Identification of Al form in the leaves. *Plant Physiol.* 113:1033–39
88. Ma JF, Ryan PR, Delhaize E. 2001. Aluminium tolerance in plants and the complexing role of organic acids. *Trends Plant Sci.* 6:273–78
89. Ma JF, Shen R, Zhao Z, Wissuwa M, Takeuchi Y, et al. 2002. Response of rice to Al stress and identification of quantitative trait loci for Al tolerance. *Plant Cell Physiol.* 43:652–59
90. Ma JF, Taketa S, Yang ZM. 2000. Aluminium tolerance genes on the short arm of chromosome 3R are linked to organic acid release in triticale. *Plant Physiol.* 122:687–94
91. Ma JF, Zheng SJ, Matsumoto H. 1997. Specific secretion of citric acid induced by Al stress in *Cassia tora* L. *Plant Cell Physiol.* 38:1019–25
92. Ma JF, Zheng SJ, Matsumoto H,

- Hiradate S. 1997. Detoxifying aluminum with buckwheat. *Nature* 390:569–70
93. Ma Z, Bielenberg D, Brown K, Lynch J. 2001. Regulation of root hair density by phosphorus availability in *Arabidopsis thaliana*. *Plant Cell Environ.* 24:459–67
94. Ma Z, Miyasaka SC. 1998. Oxalate exudation by taro in response to Al. *Plant Physiol.* 118:861–65
95. MacDiarmid CW, Gardner RC. 1996. Al toxicity in yeast. A role for Mg? *Plant Physiol.* 112:1101–9
96. MacDiarmid CW, Gardner RC. 1998. Overexpression of the *Saccharomyces cerevisiae* magnesium transport system confers resistance to aluminum ions. *J. Biol. Chem.* 273:1727–32
97. Deleted in proof
98. Magalhaes JV. 2002. Molecular genetic and physiological investigations of aluminum tolerance in sorghum (*Sorghum bicolor* L. Moench). PhD thesis. Cornell Univ. 192 pp.
99. Magnavaca R, Gardner CO, Clark RB. 1987. Inheritance of aluminum tolerance in maize. In *Genetic Aspects of Plant Mineral Nutrition*, ed. HW Gabelman, BC Loughman, pp. 201–12. Dordrecht, The Netherlands: Martinus Nijhoff
100. Matsumoto H. 2000. Cell biology of aluminum toxicity and tolerance in higher plants. *Int. Rev. Cytol.* 200:1–46
101. Matsumoto H, Hirasawa E, Morimura S, Takahashi E. 1976. Localization of aluminum in tea leaves. *Plant Cell Physiol.* 17:627–31
102. Matsumoto H, Yamamoto Y, Rama Devi S. 2001. Aluminum toxicity in acid soils. Plant responses to aluminum. In *Metals in the Environment*, ed. M Prasad, pp. 289–319. New York: Marcel Dekker
103. McLachlan K. 1980. Acid phosphatase of intact roots and phosphorus nutrition in plants. *Aust. J. Agric. Res.* 31:441–48
104. Miftahudin, Scoles GJ, Gustafson JP. 2002. AFLP markers tightly linked to the aluminum-tolerance gene Alt3 in rye (*Secale cereale* L.). *Theor. Appl. Genet.* 104:626–31
105. Milla MA, Butler E, Huete AR, Wilson CF, Anderson O, Gustafson JP. 2002. Expressed sequence tag-based gene expression analysis under aluminum stress in rye. *Plant Physiol.* 130:1706–16
106. Minella E, Sorrells ME. 1992. Aluminum tolerance in barley genetic relationships among genotypes of diverse origin. *Crop Sci.* 32:593–98
107. Miyasaka SC, Buta JG, Howell RK, Foy CD. 1991. Mechanisms of aluminum tolerance in snapbeans. Root exudation of citric acid. *Plant Physiol.* 96:737–43
108. Mullet J, Klein R, Klein P. 2001. Sorghum bicolor—an important species for comparative grass genomics and a source of beneficial genes for agriculture. *Curr. Opin. Plant Biol.* 5:118–21
109. Narang R, Altmann T. 2001. Phosphate acquisition heterosis in *Arabidopsis thaliana*: a morphological and physiological analysis. *Plant Soil* 234:91–97
110. Narang RA, Bruene A, Altmann T. 2000. Analysis of phosphate acquisition efficiency in different *Arabidopsis* accessions. *Plant Physiol.* 124:1786–99
111. Neumann G, Martinoia E. 2002. Cluster roots—an underground adaptation for survival in extreme environments. *Trends Plant Sci.* 7:162–67
112. Nguyen BD, Brar DS, Bui BC, Nguyen TV, Pham LN, Nguyen HT. 2003. Identification and mapping of the QTL for aluminum tolerance introgressed from the new source, *Oryza rufipogon* Griff., into indica rice (*Oryza sativa* L.). *Theor. Appl. Genet.* 106:583–93
113. Nguyen VT, Burow MD, Nguyen HT, Le BT, Le TD, Paterson AH. 2001. Molecular mapping of genes conferring aluminum tolerance in rice (*Oryza sativa* L.). *Theor. Appl. Genet.* 102:1002–10
114. Nguyen VT, Nguyen BD, Sarkarung S, Martinez C, Paterson AH, Nguyen HT. 2002. Mapping of genes controlling aluminum tolerance in rice: comparison

- of different genetic backgrounds. *Mol. Genet. Genomics* 267:772–80
115. Ni J, Wu P, Senadhira D, Huang N. 1998. Mapping of QTLs for phosphorus-deficiency tolerance in rice (*Oryza sativa* L.). *Theor. Appl. Genet.* 97:1361–69
116. Ninamango CFE, Guimaraes CT, Martins PR, Parentoni SN, Carneiro NP, et al. 2003. Mapping QTLs for aluminum tolerance in maize. *Euphytica* 130:223–32
117. Nürnberger T, Abel S, Jost W, Glund K. 1990. Induction of an extracellular ribonuclease in culture tomato cells upon phosphate starvation. *Plant Physiol.* 92:970–76
118. Ofei-Manu P, Wagatsuma T, Ishikawa S, Tawaraya K. 2001. The plasma membrane strength of the root-tip cells and root phenolic compounds are correlated with Al tolerance in several common woody plants. *Soil Sci. Plant Nutr.* 47:359–76
119. Osawa H, Matsumoto H. 2001. Possible involvement of protein phosphorylation in aluminum-responsive malate efflux from wheat root apex. *Plant Physiol.* 126:411–20
120. Papernik LA, Bethea AS, Singleton TE, Magalhaes JV, Garvin DF, Kochian LV. 2001. Physiological basis of reduced AL tolerance in ditelosomic lines of Chinese Spring wheat. *Planta* 212:829–34
121. Pellet DM, Grunes DL, Kochian LV. 1995. Organic acid exudation as an aluminum-tolerance mechanism in maize (*Zea mays* L.). *Planta* 196:788–95
122. Pellet DM, Papernik L, Kochian L. 1996. Multiple aluminum-resistance mechanisms in wheat. Roles of root apical phosphate and malate exudation. *Plant Physiol.* 112:591–97
123. Piñeros M, Shaff J, Kochian L. 2001. The physiology of aluminum tolerance in maize roots: Evidence that Al-activated organic acid release is not the only Al tolerance mechanism. *Plant Physiol.*, p. 97 (Abstr.)
124. Piñeros MA, Kochian LV. 2001. A patch-clamp study on the physiology of aluminum toxicity and aluminum tolerance in maize, identification and characterization of Al^{3+} -induced anion channels. *Plant Physiol.* 125:292–305
125. Piñeros MA, Magalhaes JV, Alves VMC, Kochian LV. 2002. The physiology and biophysics of an aluminum tolerance mechanism based on root citrate exudation in maize. *Plant Physiol.* 129:1194–206
126. Piper P, Mahe Y, Thompson S, Pandjaitan R, Holyoak C, et al. 1998. The Pdr12 ABC transporter is required for the development of weak organic acid resistance in yeast. *EMBO J.* 17:4257–65
127. Poirer Y, Thomas S, Sommerville C, Schiefelbein J. 1991. A mutant of *Arabidopsis* deficient in xylem loading of phosphate. *Plant Physiol.* 97:1087–93
128. Raghothama KG. 1999. Phosphate acquisition. *Annu. Rev. Plant Physiol. Plant Mol. Biol.* 50:665–93
129. Rengel Z, Zhang WH. 2003. Role of dynamics of intracellular calcium in aluminium-toxicity syndrome. *New Phytol.* 159:295–314
130. Rhue R, Grogan C, Stockmeyer E, Everett H. 1978. Genetic control of aluminum tolerance in corn. *Crop Sci.* 18:1063–67
131. Richards KD, Schott EJ, Sharma YK, Davis KR, Gardner RC. 1998. Aluminum induces oxidative stress genes in *Arabidopsis thaliana*. *Plant Physiol.* 116:409–18
132. Richards KD, Snowden KC, Gardner RC. 1994. *Wali6* and *wali7*. Genes induced by aluminum in wheat (*Triticum aestivum* L.) roots. *Plant Physiol.* 105:1455–56
133. Riede CR, Anderson JA. 1996. Linkage of RFLP markers to an aluminum tolerance gene in wheat. *Crop Sci.* 36:905–9
134. Rincon M, Gonzales R. 1992. Aluminum partitioning intact roots of aluminium-tolerant and aluminum-sensitive wheat (*Triticum aestivum* L.) cultivars. *Plant Physiol.* 99:1021–28
135. Ryan PR, Delhaize E, Jones DL. 2001. Function and mechanism of organic

- anion exudation from plant roots. *Annu. Rev. Plant Physiol. Plant Mol. Biol.* 52:527–60
136. Ryan PR, Delhaize E, Randall PJ. 1995. Characterization of Al-stimulated efflux of malate from the apices of Al-tolerant wheat roots. *Planta* 196:103–10
137. Ryan PR, Delhaize E, Randall PJ. 1995. Malate efflux from root apices and tolerance to aluminum are highly correlated in wheat. *Aust. J. Plant Physiol* 22:531–36
138. Ryan PR, DiTomaso JM, Kochian LV. 1993. Aluminum toxicity in roots: An investigation of spatial sensitivity and the role of the root cap. *J. Exp. Bot.* 44:437–46
139. Ryan PR, Skerrett M, Findlay GP, Delhaize E, Tyerman SD. 1997. Aluminum activates an anion channel in the apical cells of wheat roots. *Proc. Natl. Acad. Sci. USA* 94:6547–52
140. Saber N, Abdel-Moneim A, Barakat S. 1999. Role of organic acids in sunflower tolerance to heavy metals. *Biol. Plant.* 42:65–73
141. Sanchez P, Shepard K, Soule M, Place F, Buresh R. 1997. Soil fertility replenishment in Africa. An investment in natural resource capital. In *Replenishing Soil Fertility in Africa*, ed. R Buresh, P Sanchez, F Calhoun, pp. 1–46. Madison, WI: Soil Sci. Soc. Am.
142. Sasaki T, Yamamoto Y, Ezaki E, Katsuhara M, Ju AS, et al. 2003. A wheat gene encoding an aluminum-activated malate transporter. *Plant J.* In press
143. Schultz CJ, Rumsewicz MP, Johnson KL, Jones BJ, Gaspar YM, Bacic A. 2002. Using genomic resources to guide research directions. The arabinogalactan protein gene family as a test case. *Plant Physiol.* 129:1448–63
144. Shen R, Ma JF, Kyo M, Iwashita T. 2002. Compartmentation of aluminium in leaves of an Al-accumulator, *Fagopyrum esculentum* Moench. *Planta* 215:394–98
145. Sibov ST, Gaspar M, Silva MJ, Ottoboni LMM, Arruda P, Souza AP. 1999. Two genes control aluminum tolerance in maize: Genetic and molecular mapping analyses. *Genome* 42:475–82
146. Silva IR, Smyth TJ, Moxley DF, Carter TE, Allen NS, Rufty TW. 2000. Aluminum accumulation at nuclei of cells in the root tip. Fluorescence detection using lumogallion and confocal laser scanning microscopy. *Plant Physiol.* 123:543–52
147. Silva IR, Smyth TJ, Raper CD, Carter TE, Rufty TW. 2001. Differential aluminum tolerance in soybean: An evaluation of the role of organic acids. *Physiol Plant.* 112:200–10
148. Sivaguru M, Baluska F, Volkmann D, Felle HH, Horst WJ. 1999. Impacts of aluminum on the cytoskeleton of the maize root apex. short-term effects on the distal part of the transition zone. *Plant Physiol.* 119:1073–82
149. Sivaguru M, Ezaki B, He ZH, Tong H, Osawa H, et al. 2003. Aluminum-induced gene expression and protein localization of a cell wall-associated receptor kinase in Arabidopsis. *Plant Physiol.* 132:2256–66
150. Sivaguru M, Horst W. 1998. The distal part of the transition zone is the most aluminum-sensitive apical root zone of maize. *Plant Physiol.* 116:155–63
151. Sivaguru M, Pike S, Gassmann W, Baskin TI. 2003. Aluminum rapidly depolymerizes cortical microtubules and depolarizes the plasma membrane: evidence that these responses are mediated by a glutamate receptor. *Plant Cell Physiol.* 44:667–75
152. Snowden KC, Gardner RC. 1993. Five genes induced by aluminum in wheat (*Triticum aestivum* L.) roots. *Plant Physiol.* 103:855–61
153. Tang Y, Sorrells ME, Kochian LV, Garvin DG. 2000. Identification of RFLP markers linked to barley aluminum tolerance gene. *Alp. Crop Sci.* 40:778–82
154. Taylor GJ. 1991. Current views of the aluminum stress response: The physiological basis of tolerance. *Curr. Top. Plant Biochem. Physiol.* 10:57–93

155. Taylor GJ, McDonald-Stephens JL, Hunter DB, Bertsch PM, Elmore D, et al. 2000. Direct measurement of aluminum uptake and distribution in single cells of *Chara corallina*. *Plant Physiol.* 123: 987–96
156. Tesfaye M, Temple SJ, Allan DL, Vance CP, Samac DA. 2001. Overexpression of malate dehydrogenase in transgenic alfalfa enhances organic acid synthesis and confers tolerance to aluminum. *Plant Physiol.* 127:1836–44
157. Tice K, Parker D, DeMason D. 1992. Operationally defined apoplastic and symplastic aluminum fractions in root tips of aluminum-intoxicated wheat. *Plant Physiol.* 100:309–18
158. Trull MC, Deikman J. 1998. An Arabidopsis mutant missing one acid phosphatase isoform. *Planta* 206:544–50
159. von Uexküll HR, Mutert E. 1995. Global extent, development and economic impact of acid soils. In *Plant-Soil Interactions at Low pH: Principles and Management*, ed. RA Date, NJ Grundon, GE Raymet, ME Probert, pp. 5–19. Dordrecht, The Neth: Kluwer Academic
160. Ware DH, Jaiswal P, Ni J, Yap IV, Pan X, et al. 2002. Gramene, a tool for grass genomics. *Plant Physiol.* 130:1606–13
161. Wasaki J, Yamamura T, Shinano T, Osaki M. 2003. Secreted acid phosphatase is expressed in cluster roots of lupin in response to phosphorus deficiency. *Plant Soil* 248:129–36
162. Watt DA. 2003. Aluminium-responsive genes in sugarcane: identification and analysis of expression under oxidative stress. *J. Exp. Bot.* 54:1163–74
163. Wenzl P, Chaves A, Patino G, Mayer J, Rao I, et al. 2002. Aluminum stress stimulates the accumulation of organic acids in root apices of Brachiaria species. *J. Plant Nutr. Soil Sci.* 165:582–88
164. Wenzl P, Patino GM, Chaves AL, Mayer JE, Rao IM. 2001. The high level of aluminum resistance in signalgrass is not associated with known mechanisms of external aluminum detoxification in root apices. *Plant Physiol.* 125:1473–84
165. Williamson LC, Ribrioux SP, Fitter AH, Leyser HM. 2001. Phosphate availability regulates root system architecture in Arabidopsis. *Plant Physiol.* 126:875–82
166. Wissuwa M, Ae N. 2001. Further characterization of two QTLs that increase phosphorus uptake of rice (*Oryza sativa* L.) under phosphorus deficiency. *Plant Soil* 237:275–86
167. Wissuwa M, Yano M, Ae N. 1998. Mapping of QTLs for phosphorus-deficiency tolerance in rice (*Oryza sativa* L.). *Theor. Appl. Genet.* 97:777–83
168. Wu P, Liao CY, Hu B, Yi KK, Jin WZ, et al. 2000. QTLs and epistasis for aluminum tolerance in rice (*Oryza sativa* L.) at different seedling stages. *Theor. Appl. Genet.* 100:1295–303
169. Wu ZY, Liang F, Hong BM, Young JC, Sussman MR, et al. 2002. An endoplasmic reticulum-bound $\text{Ca}^{2+}/\text{Mn}^{2+}$ pump, ECA1, supports plant growth and confers tolerance to Mn^{2+} stress. *Plant Physiol.* 130:128–37
170. Yamamoto Y, Kobayashi Y, Devi SR, Rikiishi S, Matsumoto H. 2002. Aluminum toxicity is associated with mitochondrial dysfunction and the production of reactive oxygen species in plant cells. *Plant Physiol.* 128:63–72
171. Yamamoto Y, Kobayashi Y, Matsumoto H. 2001. Lipid peroxidation is an early symptom triggered by aluminum, but not the primary cause of elongation inhibition in pea roots. *Plant Physiol.* 125:199–208
172. Yan XL, Liao H, Trull MC, Beebe SE, Lynch JP. 2001. Induction of a major leaf acid phosphatase does not confer adaptation to low phosphorus availability in common bean. *Plant Physiol.* 125:1901–11
173. Yang Z, Sivaguru M, Horst W, Matsumoto H. 2000. Aluminium tolerance is achieved by exudation of citric acid from roots

- of soybean (*Glycine max*). *Physiol Plant*. 110:72–77
174. Zakhleniuk OV, Raines CA, Lloyd JC. 2001. *pho3*: a phosphorus-deficient mutant of *Arabidopsis thaliana* (L.) Heynh. *Planta* 212:529–34
175. Zhang WH, Rengel Z. 1999. Aluminium induces an increase in cytoplasmic calcium in intact wheat root apical cells. *Aust. J. Plant Physiol*. 26:401–9
176. Zhang WH, Ryan PR, Tyerman SD. 2001. Malate-permeable channels and cation channels activated by aluminum in the apical cells of wheat roots. *Plant Physiol*. 125:1459–72
177. Zheng SJ, Ma JF, Matsumoto H. 1998. Continuous secretion of organic acids is related to aluminum resistance during relatively long-term exposure to aluminum stress. *Physiol. Plant*. 103:209–14
178. Zheng SJ, Ma JF, Matsumoto H. 1998. High aluminum resistance in buckwheat: I. Al-induced specific secretion of oxalic acid from root tips. *Plant Physiol*. 117:745–51

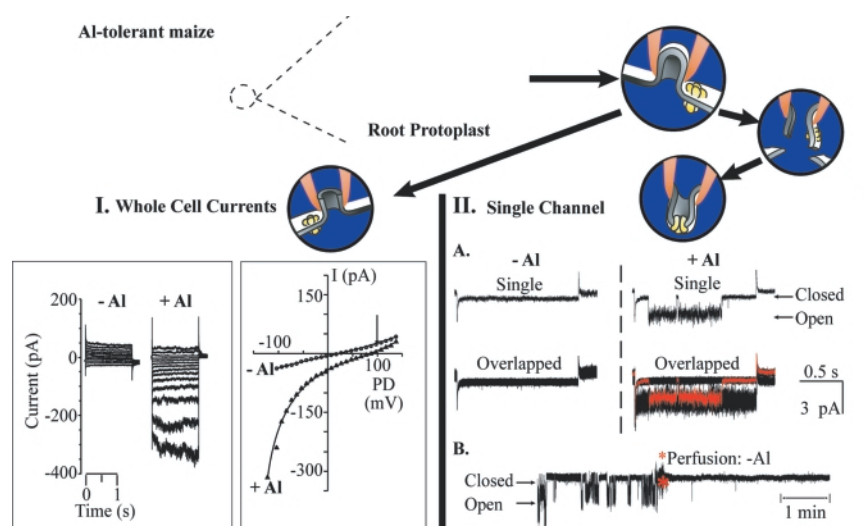


Figure 1 A model depicting Al-activated anion transporters in the plasma membrane of root cells from Al-tolerant maize. The patch clamp technique was employed to record macroscopic currents (whole-cell currents on the *left*) or to study the transporter activity in isolated membrane patches (single-channel currents on the *right*). (I) Al activates an inward plasma membrane whole-cell anion current (anion efflux). Whole-cell currents were elicited at holding potentials clamped in 10-mV increments. The bath contained 1 mM Cl^- (pH 4.0) minus (*left traces*) or plus (*right traces*) 50 μM Al^{3+} . (*Right panel*) Current-voltage relationship for the currents shown on the left. The arrow indicates the Cl^- theoretical reversal potential. (II A) Al can activate single anion channels in excised membrane patches. A negative voltage potential was employed to test for single-channel activity in outside-out membrane patches excised in the absence of extracellular Al^{3+} [the resulting single trace is the top trace in (II A)]. This voltage protocol was repeated 60 times to confirm the lack of single-channel activity in the absence of Al^{3+} (the resulting 60 traces are overlapped). Subsequently, the membrane patch was exposed to extracellular Al^{3+} (50 μM) and the same voltage protocol was employed. The single trace in the right panel shows a single sweep after Al^{3+} exposure. The resulting 60 traces obtained in the presence of Al^{3+} are shown overlapped in the bottom trace of the right panel, with the single trace from above highlighted in red. (II B) Al^{3+} is required to maintain channel activity. A continuous 6-minute single-channel trace from an outside-out excised patch showing channel activity is depicted when the bath solution contained 50 μM Al^{3+} . The asterisk indicates the time (2 min 34 sec) when the recording chamber was perfused with an identical bath solution lacking Al^{3+} . Arrows on the right indicate the closed and open states of the channel.

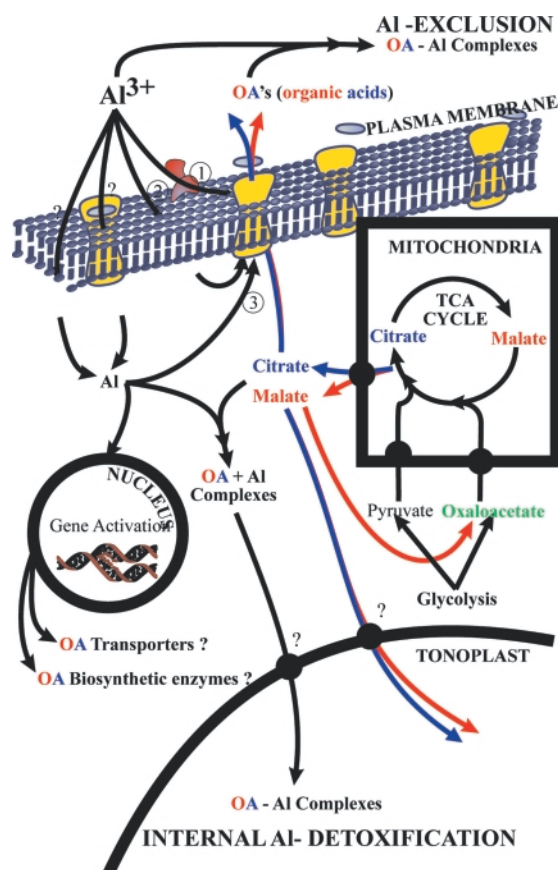


Figure 2 Model illustrating Al exclusion and Al internal detoxification mechanism based on the formation of Al complexes with organic acids (OAs). The Al exclusion mechanism involves the release of OAs via an Al gated channel at the plasma membrane. Activation of the channel could occur either by: (1) Al interacting directly with the channel or a receptor in close proximity to the channel; (2) Al interacting with a membrane-bound receptor, triggering a signal transduction pathway that results in channel activation; (3) Al entering the cytosol and initiating a signal transduction cascade involving cytoplasmic components, resulting in channel activation. Al may also trigger changes in expression of genes involved in Al tolerance. The internal Al detoxification mechanism involves chelation of cytoplasmic Al by OAs with the subsequent sequestration into the vacuole via unknown transporters.

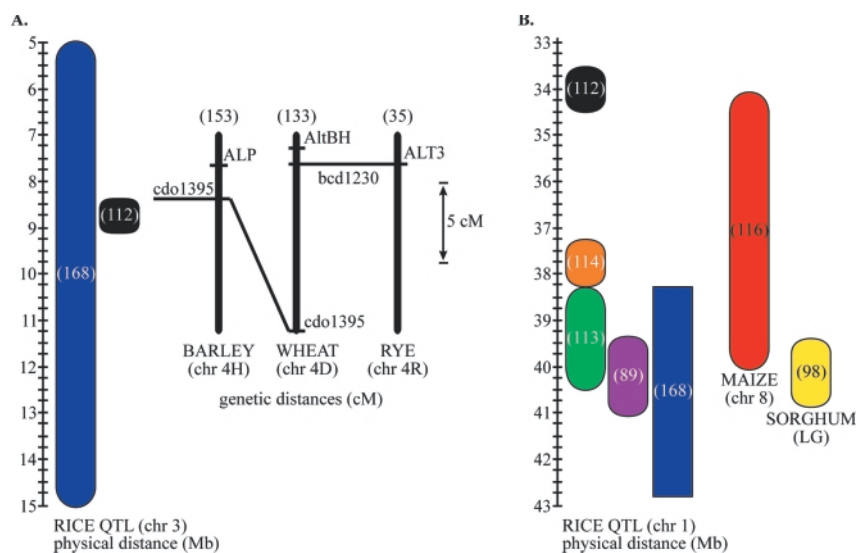


Figure 3 Comparative mapping of Al tolerance loci between grasses. (A) Loci located to rice linkage block 3C. Flanking genetic markers for the QTLs identified in rice (112, 168) were located on the rice chromosome 3 pseudomolecule using Gramene. The cdo1395 marker defined the boundary for one rice QTL (112) and serves to link rice to the Triticeae. Genetic distances between linked markers and Al tolerance genes in the Triticeae (35, 133, 153) are shown. The 5-cM scale was drawn using the physical-to-genetic distance ratio from rice over this interval (approx. 0.36 Mb/1 cM). (B) Loci located to rice linkage block 1B (34). Flanking genetic markers for the QTLs identified in rice (89, 112–114, 168), maize (116), and sorghum (98) were located on the rice chromosome 1 pseudomolecule using Gramene (160). QTL confidence intervals are indicated by the colored bars, and the numbers within parentheses on or above these colored bars denote the reference for the associated study; distances are in Mb.